

# UVW Signature Scheme

## Algorithm Description Document

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# 1 Algorithm Description

## 1.1 Notations

$\mathbb{F}_q$  denotes a finite field of size  $q$ . Bold lowercase letters (e.g.,  $\mathbf{x}$ ) and uppercase letters (e.g.,  $\mathbf{H}$ ) denote row vectors and matrices, respectively.  $\mathbf{x}^\top$  ( $\mathbf{H}^\top$ ) denotes the transpose of vector  $\mathbf{x}$  (matrix  $\mathbf{H}$ ). Let  $m, n > m$  be positive integers.  $[n]$  and  $[m, n]$  denote the integer sets  $\{1, 2, \dots, n\}$  and  $\{m, m+1, \dots, n\}$ , respectively.  $S_n$  denotes the symmetric group of permutations of degree  $n$ , and  $\mathcal{DP}(n)$  denotes the set of all monomial matrices (products of permutation matrices and invertible diagonal matrices) of order  $n$  over the finite field  $\mathbb{F}_q$ . For  $\mathbf{x} \in \mathbb{F}_q^n$ ,  $\mathbf{x}(i)$  denotes the  $i$ -th component of  $\mathbf{x}$ , sometimes also denoted as  $x_i$ , i.e.,  $\mathbf{x} = (x_i)_{i \in [n]}$ .  $|\mathbf{x}|$  denotes the Hamming weight of  $\mathbf{x}$ , i.e.,  $|\mathbf{x}| = |\{i \in [n] \mid \mathbf{x}(i) \neq 0\}|$ , sometimes also denoted as  $wt(\mathbf{x})$ .  $\lfloor a \rfloor$  denotes the floor function of a rational number  $a$ .  $\text{negl}(\lambda)$  denotes a negligible function with respect to  $\lambda$ .  $\mathcal{E}_{n,w}$  denotes the set of all vectors in the vector space  $\mathbb{F}_q^n$  with a Hamming weight of  $w$ . The size of the set is  $\binom{n}{w}(q-1)^w$ .  $s \xleftarrow{\$} S$  denotes that the element  $s$  is chosen uniformly at random from the set  $S$ . Let  $X$  be a random variable.  $\mathbb{P}(X = A)$  denotes the probability of event  $A$  occurring, and  $\mathbb{E}(X)$  is its expectation.

## 1.2 Coding Theory Basics

An  $[n, k]$  linear code  $\mathcal{C}$  over the finite field  $\mathbb{F}_q$  can be viewed as a  $k$ -dimensional subspace of the  $n$ -dimensional vector space  $\mathbb{F}_q^n$ . The vectors in the code  $\mathcal{C}$  are referred to as codewords, where  $n$  denotes the length of the code, and  $k$  represents its dimension. Analogous to representing a linear space with a set of basis vectors, a linear code  $\mathcal{C}$  can be specified by a basis comprising its codewords. The matrix formed by these basis vectors is referred to as the generator matrix of the linear code  $\mathcal{C}$  [1].

**Definition 1** (Generator Matrix and Parity-Check Matrix [2]). *Let  $\mathcal{C}$  be an  $[n, k]$  linear code over the finite field  $\mathbb{F}_q$ . A generator matrix of the code  $\mathcal{C}$  is a full-rank  $k \times n$  matrix over  $\mathbb{F}_q$ , whose row vectors form a basis for  $\mathcal{C}$ . If*

$$\mathcal{C} = \mathbf{xG} \mid \mathbf{x} \in \mathbb{F}_q^k,$$

*then  $\mathbf{G}$  is a generator matrix of the code  $\mathcal{C}$ . A parity-check matrix of the code  $\mathcal{C}$  is a full-rank  $(n - k) \times n$  matrix over  $\mathbb{F}_q$ . If*

$$\forall \mathbf{x} \in \mathcal{C} \Leftrightarrow \mathbf{xH}^\top = \mathbf{0},$$

*then  $\mathbf{H}$  is a parity-check matrix of the code  $\mathcal{C}$ .*

The generator matrix and parity-check matrix of a code  $\mathcal{C}$  are not unique. For any generator matrix  $\mathbf{G}$  and parity-check matrix  $\mathbf{H}$  of the code  $\mathcal{C}$ , it holds that  $\mathbf{GH}^\top = \mathbf{0}$ . An information set of an  $[n, k]$  linear code  $\mathcal{C}$  is a set of  $k$  coordinate indices, denoted as  $I \subset [n]$ , which corresponds to the indices of  $k$  linearly independent columns in the generator matrix  $\mathbf{G}$ . Its complement corresponds to the indices of  $r$  ( $r = n - k$ ) linearly independent columns in the parity-check matrix  $\mathbf{H}$ . Given a parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{r \times n}$  of the code  $\mathcal{C}$  and an information set  $I$ , for any vector  $\mathbf{s} \in \mathbb{F}_q^r$  and  $\mathbf{x} \in \mathbb{F}_q^n$ , there exists a unique vector  $\mathbf{e} \in \mathbb{F}_q^n$  such that  $\mathbf{eH}^\top = \mathbf{s}$  and  $\mathbf{x}_I = \mathbf{e}_I$ .

In code-based cryptographic designs, linear codes endowed with specific algebraic structures are frequently employed as the underlying trapdoors for constructing encryption or signature schemes. Among these, the  $(U, U+V)$  code, originally introduced by Plotkin [3], has emerged as a fundamental building block for numerous classical schemes due to its unique recursive construction logic and highly efficient decoding properties.

**Definition 2** ( $(U, U+V)$  Code). *A  $(U, U+V)$  code over the finite field  $\mathbb{F}_q$  is constructed from two linear codes  $U$  and  $V$ . Let  $n, k_1$ , and  $k_2$  be three positive integers, where  $n$  is even. Let  $U$  be an  $[\frac{n}{2}, k_1]$  linear code and  $V$  be an  $[\frac{n}{2}, k_2]$  linear code. Then, the generator matrix of the  $(U, U+V)$  code is defined as follows:*

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{0} & \mathbf{G}_V \end{pmatrix},$$

where  $\mathbf{G}_U$  and  $\mathbf{G}_V$  are the generator matrices of  $U$  and  $V$ , respectively.

However, as illustrated in the aforementioned definition, the standard generator matrix of a  $(U, U+V)$  code contains an explicit zero matrix block. When this unaltered structure is directly employed to construct public-key trapdoors (such as in the early SURF signature scheme [4]), it is highly susceptible to structural recovery attacks. Adversaries can readily exploit this zero block to isolate and break the underlying subcodes. To overcome this inherent vulnerability, state-of-the-art post-quantum signature schemes (e.g., the Wave scheme [5]) have introduced the concept of generalized  $(U, U+V)$  codes, which conceal the subcode structure through algebraic coupling utilizing secret vector multipliers.

Inspired by this algebraic evolution, and to further decouple the logic of the underlying subcodes from simple scalar or vector couplings, thereby more thoroughly mitigating potential structural vulnerabilities and expanding the key space, we propose a  $(U+V, U+W)$  code based on the traditional  $(U, U+V)$  code, featuring a more intricate structure and a higher algebraic degree of freedom. Below, we present the formal definition of the  $(U+V, U+W)$  code.

**Definition 3** ( $(U+V, U+W)$  Code). *A  $(U+V, U+W)$  code over the finite field  $\mathbb{F}_q$  is composed of three linear codes  $U, V$ , and  $W$ . Let  $n, k_1$ , and  $k_2$  be three positive integers, where  $n$  is even. Let  $U$  be an  $[\frac{n}{2}, k_1]$  linear code, and both  $V$  and  $W$  be  $[\frac{n}{2}, k_2]$  linear codes.  $\mathbf{G}_U, \mathbf{G}_V$ , and  $\mathbf{G}_W$  are the generator matrices of  $U, V$ , and  $W$ , respectively, and  $\mathbf{G}_V - \mathbf{G}_W$  is a full-rank matrix. The generator matrix is defined as follows:*

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix},$$

Then the  $[n, k]$  linear code ( $k = k_1 + k_2$ ) generated by  $\mathbf{G}$  is called a  $(U+V, U+W)$  code.

Next, we provide the definition of the generalized  $(U+V, U+W)$  code, which is the code used in the signature scheme of this paper.

**Definition 4** (Generalized  $(U+V, U+W)$  Code). *A generalized  $(U+V, U+W)$  code over the finite field  $\mathbb{F}_q$  is composed of three linear codes  $U, V$ , and  $W$ . Let  $n, k_1$ , and  $k_2$  be three positive*

integers, where  $n$  is even. Let  $U$  be an  $[\frac{n}{2}, k_1]$  linear code, and both  $V$  and  $W$  be  $[\frac{n}{2}, k_2]$  linear codes.  $\mathbf{D} \in \mathbb{F}_q^{n \times n}$  is a random monomial matrix (which can be represented as the product of a permutation matrix and an invertible diagonal matrix).  $\mathbf{G}_U, \mathbf{G}_V$ , and  $\mathbf{G}_W$  are the generator matrices of  $U, V$ , and  $W$ , respectively, and  $\mathbf{G}_V - \mathbf{G}_W$  is a full-rank matrix. The generator matrix is defined as follows:

$$\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \mathbf{D},$$

Then the  $[n, k]$  linear code ( $k = k_1 + k_2$ ) generated by  $\mathbf{G}$  is called a generalized  $(U + V, U + W)$  code.

**Corollary 1.** Let  $\mathcal{C}$  be a generalized  $[n, k]$   $(U + V, U + W)$  code with the generator matrix  $\mathbf{G} = \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \mathbf{D}$ . Define its dual code  $\mathcal{C}^\perp$  as

$$\mathcal{C}^\perp = \{\mathbf{y} \in \mathbb{F}_q^n \mid \forall \mathbf{x} \in \mathcal{C}, \langle \mathbf{x}, \mathbf{y} \rangle = 0\},$$

Then, the dual code  $\mathcal{C}^\perp$  is a generalized  $[n, n - k]$   $(U + V, U + W)$  code, and there exists a structured parity-check matrix for it.

**Proof.** From Definition 4,  $\mathbf{G}_U, \mathbf{G}_V$ , and  $\mathbf{G}_W$  are the generator matrices of  $U, V$ , and  $W$ , respectively, and  $\mathbf{G}_\Delta = \mathbf{G}_W - \mathbf{G}_V$  is of full row rank. To construct the parity-check matrix  $\mathbf{H}$  of  $\mathcal{C}$  (i.e., the generator matrix of the dual code  $\mathcal{C}^\perp$ ), we define the following three matrices directly based on the duality of the underlying subcodes: 1) Let  $\mathbf{H}_U \in \mathbb{F}_q^{(n/2 - k_1) \times n/2}$  be the parity-check matrix of subcode  $U$ , satisfying  $\mathbf{G}_U \mathbf{H}_U^T = \mathbf{0}$ ; 2) Let  $\mathbf{H}_\Delta \in \mathbb{F}_q^{(n/2 - k_2) \times n/2}$  be the parity-check matrix of the subcode generated by  $\mathbf{G}_\Delta$ , satisfying  $(\mathbf{G}_W - \mathbf{G}_V) \mathbf{H}_\Delta^T = \mathbf{0}$ ; 3) Since  $\mathbf{G}_\Delta$  is of full row rank, the non-homogeneous system of linear equations  $\mathbf{G}_\Delta \mathbf{Y}^T = -\mathbf{G}_V \mathbf{H}_U^T$  has infinitely many solutions. Relying on the degrees of freedom in the solution space, we can select a specific particular solution matrix denoted as  $\mathbf{H}_Y \in \mathbb{F}_q^{(n/2 - k_1) \times n/2}$  such that  $\mathbf{H}_Y$  itself is of full rank, and the derived matrix difference  $\mathbf{H}_U - \mathbf{H}_Y$  is also a full-rank matrix.

Based on the components above, we construct the candidate parity-check matrix  $\mathbf{H}$  as follows:

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_U - \mathbf{H}_Y & \mathbf{H}_Y \\ -\mathbf{H}_\Delta & \mathbf{H}_\Delta \end{pmatrix} (\mathbf{D}^{-1})^T.$$

We need to verify  $\mathbf{G} \mathbf{H}^T = \mathbf{0}$ . Expanding the matrix multiplication:

$$\begin{aligned} \mathbf{G} \mathbf{H}^T &= \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \mathbf{D} \cdot \mathbf{D}^{-1} \begin{pmatrix} (\mathbf{H}_U - \mathbf{H}_Y)^T & -\mathbf{H}_\Delta^T \\ \mathbf{H}_Y^T & \mathbf{H}_\Delta^T \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{G}_U & \mathbf{G}_U \\ \mathbf{G}_V & \mathbf{G}_W \end{pmatrix} \begin{pmatrix} \mathbf{H}_U^T - \mathbf{H}_Y^T & -\mathbf{H}_\Delta^T \\ \mathbf{H}_Y^T & \mathbf{H}_\Delta^T \end{pmatrix} \end{aligned}$$

Calculate the four block matrices of the resulting matrix respectively:

1. Top-left block:  $\mathbf{G}_U(\mathbf{H}_U^T - \mathbf{H}_Y^T) + \mathbf{G}_U \mathbf{H}_Y^T = \mathbf{G}_U \mathbf{H}_U^T - \mathbf{G}_U \mathbf{H}_Y^T + \mathbf{G}_U \mathbf{H}_Y^T = \mathbf{G}_U \mathbf{H}_U^T = \mathbf{0}$ ;

2. Top-right block:  $-\mathbf{G}_U \mathbf{H}_\Delta^T + \mathbf{G}_U \mathbf{H}_\Delta^T = \mathbf{0}$ ;
3. Bottom-right block:  $-\mathbf{G}_V \mathbf{H}_\Delta^T + \mathbf{G}_W \mathbf{H}_\Delta^T = \mathbf{G}_\Delta \mathbf{H}_\Delta^T = \mathbf{0}$ ;
4. Bottom-left block:  $\mathbf{G}_V(\mathbf{H}_U^T - \mathbf{H}_Y^T) + \mathbf{G}_W \mathbf{H}_Y^T = \mathbf{G}_V \mathbf{H}_U^T + (\mathbf{G}_W - \mathbf{G}_V) \mathbf{H}_Y^T$ . Substituting the definition of  $\mathbf{H}_Y$ , which is  $(\mathbf{G}_W - \mathbf{G}_V) \mathbf{H}_Y^T = -\mathbf{G}_V \mathbf{H}_U^T$ , we obtain  $\mathbf{G}_V \mathbf{H}_U^T - \mathbf{G}_V \mathbf{H}_U^T = \mathbf{0}$ . Thus,  $\mathbf{G} \mathbf{H}^T = \mathbf{0}$  is strictly proven.

Next, we verify the rank of the parity-check matrix  $\mathbf{H}$ . Consider the core block matrix  $\mathbf{M} = \begin{pmatrix} \mathbf{H}_U - \mathbf{H}_Y & \mathbf{H}_Y \\ -\mathbf{H}_\Delta & \mathbf{H}_\Delta \end{pmatrix}$  of  $\mathbf{H}$ . By adding the right column block of  $\mathbf{M}$  to the left column block (elementary column operations do not change the rank of a matrix), we obtain:

$$\mathbf{M} \xrightarrow{C_1 \leftarrow C_1 + C_2} \begin{pmatrix} (\mathbf{H}_U - \mathbf{H}_Y) + \mathbf{H}_Y & \mathbf{H}_Y \\ -\mathbf{H}_\Delta + \mathbf{H}_\Delta & \mathbf{H}_\Delta \end{pmatrix} = \begin{pmatrix} \mathbf{H}_U & \mathbf{H}_Y \\ \mathbf{0} & \mathbf{H}_\Delta \end{pmatrix}.$$

This transforms into a block upper triangular matrix. The rank of this matrix is the sum of the ranks of its diagonal blocks:

$$\text{Rank}(\mathbf{M}) = \text{Rank}(\mathbf{H}_U) + \text{Rank}(\mathbf{H}_\Delta) = (n/2 - k_1) + (n/2 - k_2) = n - (k_1 + k_2) = n - k.$$

Since  $(\mathbf{D}^{-1})^T$  is an invertible matrix, right-multiplying it does not change the rank. Therefore,  $\mathbf{H} = \mathbf{M}(\mathbf{D}^{-1})^T$  has full rank  $n - k$ .

By adjusting the block row order of matrix  $\mathbf{H}$  and absorbing the negative sign (diagonal permutation), it can be transformed into the isomorphic form of the generator matrix of a standard generalized  $(U + V, U + W)$  code. Therefore, the dual code  $\mathcal{C}^\perp$  is also a generalized  $(U + V, U + W)$  code.  $\square$

To describe the matrix block operations more intuitively in the subsequent signature decoding algorithm, we define  $\mathbf{H}_X = \mathbf{H}_U - \mathbf{H}_Y$  and  $\mathbf{H}_Z = \mathbf{H}_\Delta$ . Concurrently, it is known that  $(\mathbf{D}^{-1})^T$  remains a monomial matrix. Thus, the parity-check matrix can be equivalently rewritten as:

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D}$$

From Definition 1, any linear code can be represented by a generator matrix  $\mathbf{G}$  or a parity-check matrix  $\mathbf{H}$ . Therefore, we use the parity-check matrix defined above to represent a generalized  $[n, k]$   $(U + V, U + W)$  code, where  $\mathbf{H}_X, \mathbf{H}_Y \in \mathbb{F}_q^{(n/2 - k_1) \times n/2}$  (with rank  $n/2 - k_1$ ),  $\mathbf{H}_Z \in \mathbb{F}_q^{(n/2 - k_2) \times n/2}$  (with rank  $n/2 - k_2$ ),  $\mathbf{H}_X + \mathbf{H}_Y$  is a full-rank matrix,  $k = k_1 + k_2$ , and  $\mathbf{D} \in \mathbb{F}_q^{n \times n}$  is a random monomial matrix.

### 1.3 Auxiliary Functions

This section presents the fundamental algebraic subroutines relied upon by the UVW signature scheme during key generation, signature generation, and verification. Among these, **InfSet**,

PrangeStep, and Prange constitute the foundational algorithmic engine based on syndrome decoding;  $\text{Dec}_Z$  and  $\text{Dec}_{XY}$  are specialized trapdoor decoding subroutines designed for the generalized  $(U + V, U + W)$  code structure.

**Information Set Selection.** Information Set Decoding (ISD), as the core technique for solving the syndrome decoding problem, was first proposed by Prange in 1962 [6]. Over the subsequent decades, researchers such as Stern [7] and Dumer [8] greatly extended it by introducing the birthday paradox and collision search, forming the fundamental framework of modern ISD. Although modern ISD algorithms have become increasingly complex, the information set extraction method defined in Algorithm 1—namely, extracting an information set  $\mathcal{I}$  of size  $k$  from a given parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$ , thereby transforming the global error search into solving a local system of linear equations—remains an indispensable first step for all advanced algorithms.

---

**Algorithm 1** InfSet( $\mathbf{H}$ )

---

**Input:** Matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$

**Output:** An information set  $\mathcal{I}$  of size  $k$ , whose complement corresponds to  $n - k$  linearly independent columns of the parity-check matrix

---

**Prange Iteration Subroutine.** Algorithm 2 defines the standard single-step algebraic operation for syndrome decoding. Given an information set  $\mathcal{I}$  and a predefined target error vector pattern  $\mathbf{x}$ , the algorithm uses a permutation matrix  $\mathbf{P}$  to move the columns corresponding to the information set to the end. It then utilizes the block matrix  $(\mathbf{A}||\mathbf{B})$  to solve the non-homogeneous system of linear equations, deterministically constructing an error vector  $\mathbf{e}$  that satisfies the syndrome constraint  $\mathbf{eH}^\top = \mathbf{s}$ . This block-solving technique based on Gaussian elimination is not only efficient but also serves as a crucial entry point for implementing constant-time side-channel protection at the lowest level [9].

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**Algorithm 2** PrangeStep( $\mathbf{H}, \mathbf{s}, \mathcal{I}, \mathbf{x}$ )

---

**Input:** Matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$ , vector  $\mathbf{s} \in \mathbb{F}_q^{n-k}$ , information set  $\mathcal{I}$ , and vector  $\mathbf{x} \in \mathbb{F}_q^n$

- 1: Choose a permutation matrix  $\mathbf{P} \in \mathbb{F}_q^{n \times n}$  such that  $\mathcal{I}$  is in the last  $k$  coordinates;
- 2:  $(\mathbf{A}||\mathbf{B}) \leftarrow \mathbf{HP}$ ,  $(\mathbf{0}||\mathbf{e}'') \leftarrow \mathbf{xP}$ ;
- 3:  $\mathbf{e} \leftarrow ((\mathbf{s} - \mathbf{e}''\mathbf{B}^\top)(\mathbf{A}^{-1})^\top || \mathbf{e}'')\mathbf{P}^\top$ ;

**Output:**  $\mathbf{e}$

---

**Prange Decoding Algorithm.** Algorithm 3 defines the complete process of large-weight syndrome decoding for a specific Hamming weight  $w$ . Based on the target weight  $w$  and the characteristics of the finite field, the algorithm first intelligently computes the optimal number of non-zero elements  $t$  distributed over the information set. Subsequently, it generates a matching vector pattern  $\mathbf{x}$  and invokes the aforementioned information set selection and single-step iteration subroutines to output, with exceptionally high probability, an error vector strictly of weight  $w$ .

---

**Algorithm 3** Prange( $\mathbf{H}, \mathbf{s}, w$ )

---

**Input:** Matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$ , vector  $\mathbf{s} \in \mathbb{F}_q^{n-k}$ , and integer  $w$

- 1:  $t \leftarrow \min\left(k, \max\left(0, \left\lceil w - \frac{q-1}{q}(n-k) \right\rceil\right)\right)$ ;
- 2:  $\mathcal{I} \leftarrow \text{InfSet}(\mathbf{H})$ ;
- 3:  $\mathbf{x} \leftarrow \{\mathbf{x} \in \mathbb{F}_q^n \mid |\mathbf{x}| = |\mathbf{x}_{\mathcal{I}}| = t\}$ ;
- 4:  $\mathbf{e} \leftarrow \text{PrangeStep}(\mathbf{H}, \mathbf{s}, \mathcal{I}, \mathbf{x})$ ;

**Output:**  $\mathbf{e}$

---

**Z-Domain and XY-Domain Decoding Subroutines.** Algorithm 4 defines the local decoding process targeting the underlying subcode parity-check matrix  $\mathbf{H}_Z$ . By introducing a specific distribution  $\mathcal{D}_2$  to control the weight  $t$  of the local error vector, it invokes **PrangeStep** to deterministically compute a local error vector  $\mathbf{e}_2$  that satisfies the equation  $\mathbf{e}_2 \mathbf{H}_Z^T = \mathbf{s}_2$ . Subsequently, based on the known local error  $\mathbf{e}_2$ , Algorithm 5 solves the correlated equation  $\mathbf{e}_1 \mathbf{H}_{XY}^T = \mathbf{s}'_1$ . To ensure that the total Hamming weight of the final signature strictly equals the target value  $w$ , this algorithm incorporates a rigorous rejection sampling loop, continuously retrying by selecting new information sets until the total weight matches perfectly. This design not only ensures that the statistical properties of the final signature vector are indistinguishable from those of a purely random large-weight vector [10], but also achieves a complete statistical decoupling of the signature generation time from the private key structure, thereby immunizing against side-channel leakage risks at the algorithmic level.

---

**Algorithm 4** Dec<sub>Z</sub>( $\mathbf{H}_Z, \mathbf{s}_2$ )

---

**Input:** Matrix  $\mathbf{H}_Z \in \mathbb{F}_3^{(n/2-k_2) \times n/2}$ , vector  $\mathbf{s}_2 \in \mathbb{F}_3^{n/2-k_2}$ , and a distribution  $\mathcal{D}_2$  over the set  $[0, k_2]$

- 1:  $t \leftarrow \mathcal{D}_2$ ;
- 2:  $\mathcal{I} \xleftarrow{\$} \text{InfSet}(\mathbf{H}_Z)$ ;
- 3:  $\mathbf{x} \xleftarrow{\$} \{\mathbf{x} \in \mathbb{F}_3^{n/2} \mid |\mathbf{x}| = |\mathbf{x}_{\mathcal{I}}| = t\}$ ;
- 4:  $\mathbf{e}_2 \leftarrow \text{PrangeStep}(\mathbf{H}_Z, \mathbf{s}_2, \mathcal{I}, \mathbf{x})$ ;

**Output:**  $\mathbf{e}_2$

---



---

**Algorithm 5** Dec<sub>XY</sub>( $\mathbf{H}_{XY}, \mathbf{s}'_1, \mathbf{e}_2$ )

---

**Input:** Matrix  $\mathbf{H}_{XY} \in \mathbb{F}_3^{(n/2-k_1) \times n/2}$ , vector  $\mathbf{s}'_1 \in \mathbb{F}_3^{n/2-k_1}$ , and  $\mathbf{e}_2 \in \mathbb{F}_3^{n/2}$

- 1:  $\mathcal{I} \xleftarrow{\$} \text{InfSet}(\mathbf{H}_{XY})$ ;
- 2: **repeat**
- 3:    $\mathbf{x} \xleftarrow{\$} \{\mathbf{x} \in \mathbb{F}_3^{n/2} \mid |\mathbf{x}| = |\mathbf{x}_{\mathcal{I}}| = |(\mathbf{x} + \mathbf{e}_2)_{\mathcal{I}}| = k_1\}$ ;
- 4:    $\mathbf{e}_1 \leftarrow \text{PrangeStep}(\mathbf{H}_{XY}, \mathbf{s}'_1, \mathcal{I}, \mathbf{x})$ ;
- 5: **until**  $|(\mathbf{e}_1 | \mathbf{e}_1 + \mathbf{e}_2)| = w$

**Output:**  $\mathbf{e}_1$

---

## 1.4 Overview of the UVW Signature Scheme

The UVW signature scheme consists of three components: key generation, signature generation, and verification algorithms. Figure 1 provides a brief description of the UVW scheme.

From the description below, the signature size of the scheme is:  $|\sigma| = \lambda + k \log_2(q)$ . During the public key generation, transforming the parity-check matrix into systematic form  $(\mathbf{I}_{n-k} || \mathbf{R})$  via elementary row operations is equivalent to left-multiplying by  $\mathbf{H}_1^{-1}$ . Therefore, the public and private keys of the scheme are defined as follows:

$$\text{pk} : \mathbf{H}_{\text{pk}} = \mathbf{H}_1^{-1} \mathbf{H}_{\text{sk}} = (\mathbf{I}_{n-k} || \mathbf{R}), \text{ sk} : (\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z, \mathbf{D}).$$

**Key Generation.**  $\text{KeyGen}(1^\lambda)$ : Taking the security parameter  $\lambda$  as input, randomly generate full-rank matrices  $\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z$  and a monomial matrix  $\mathbf{D}$ , such that the square matrix  $\mathbf{H}_1$ , consisting of the first  $n - k$  columns of  $\mathbf{H}_{\text{sk}}$ , is invertible. Output the public key  $\text{pk} : \mathbf{H}_{\text{pk}} = (\mathbf{I}_{n-k} || \mathbf{R})$ , the private key  $\text{sk} : (\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z, \mathbf{D})$ , and the public parameters  $\text{params} = (q, n, k, k_1, k_2, w)$ , where  $k = k_1 + k_2$ . In addition, a secure hash function  $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{F}_q^{n-k}$  is required.

**Signature.**  $\text{Sign}(\text{sk}, \mathbf{m})$ : Taking the private key  $\text{sk}$  and a message  $\mathbf{m} \in \{0, 1\}^*$  as input, output the signature  $\sigma = (\mathbf{e}, \mathbf{r}) \in \mathbb{F}_q^k \times \mathbb{F}_2^\lambda$ .

- 1: Choose a random vector  $\mathbf{r} \xleftarrow{\$} \{0, 1\}^\lambda$
- 2: Compute  $\mathbf{s} = \mathcal{H}(\mathbf{m}, \mathbf{r})$
- 3: Reconstruct the parity-check matrix  $\mathbf{H}_{\text{sk}}$  and extract its first  $n - k$  columns to obtain the square matrix  $\mathbf{H}_1$ ; compute  $(\mathbf{s}_1 || \mathbf{s}_2) = \mathbf{s} \mathbf{H}_1^\top$ , where  $\mathbf{s}_1 \in \mathbb{F}_q^{n/2-k_1}, \mathbf{s}_2 \in \mathbb{F}_q^{n/2-k_2}$
- 4:  $\mathcal{I} \xleftarrow{\$} \text{InfSet}(\mathbf{H}_Z)$
- 5:  $\mathbf{e}_2 \leftarrow \text{PrangeStep}(\mathbf{H}_Z, \mathbf{s}_2, \mathcal{I}, \mathbf{0})$
- 6:  $\mathcal{I} \xleftarrow{\$} \text{InfSet}(\mathbf{H}_X + \mathbf{H}_Y)$
- 7:  $\mathbf{x} \xleftarrow{\$} \{\mathbf{x} \in \mathbb{F}_q^{n/2} \mid |\mathbf{x}| = |\mathbf{x}_{\mathcal{I}}| = |(\mathbf{x} + \mathbf{e}_2)_{\mathcal{I}}| = t\}$
- 8:  $\mathbf{H}_{XY} = \mathbf{H}_X + \mathbf{H}_Y, \mathbf{s}'_1 = \mathbf{s}_1 - \mathbf{e}_2 \mathbf{H}_Y^\top$
- 9:  $\mathbf{e}_1 \leftarrow \text{PrangeStep}(\mathbf{H}_{XY}, \mathbf{s}'_1, \mathcal{I}, \mathbf{x})$ ; if  $|(\mathbf{e}_1 || \mathbf{e}_1 + \mathbf{e}_2)| = w$ , continue; otherwise, return to Step 7
- 10: Compute  $(\mathbf{e}' || \mathbf{e}) = (\mathbf{e}_1 || \mathbf{e}_1 + \mathbf{e}_2)(\mathbf{D}^\top)^{-1}$ , where  $\mathbf{e}', \mathbf{e} \in \mathbb{F}_q^{n/2}$
- 11: Output  $\sigma = (\mathbf{e}, \mathbf{r})$

**Verification.**  $\text{Verify}(\text{pk}, \sigma)$ : Taking the public key  $\text{pk}$  and the signature  $\sigma$  as input, verify whether the equation  $|\mathbf{e}| + |\mathcal{H}(\mathbf{m}, \mathbf{r}) - \mathbf{e} \mathbf{R}^\top| = w$  holds.

Figure 1: Algorithmic description of the UVW signature scheme

## 1.5 Specific Construction of the UVW Signature Scheme

The UVW signature scheme is constructed based on generalized  $(U + V, U + W)$  codes. The scheme includes 6 integer parameters  $(q, n, k, k_1, k_2, w)$ , a secure hash function  $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{F}_3^{n-k}$ , and a pseudo-random number generator  $\text{PRNG} : \{0, 1\}^* \rightarrow \{0, 1\}^*$  for deterministic private key expansion (its output bitstream can be deterministically parsed into matrices over the finite field

$\mathbb{F}_3$ ). The entire scheme is defined over the finite field  $\mathbb{F}_3$ , and all subsequent algebraic descriptions are performed over  $\mathbb{F}_3$ . Section 1.6 will present the recommended parameter sets for this scheme.

To meet the engineering requirements for extremely low storage overhead in practical applications, the specific construction of the UVW signature scheme presented in this section adopts a private key compression strategy. This construction compresses the private key size to the extreme by reusing a high-entropy master seed and introducing short salt values for deterministic expansion. Algorithm 6, Algorithm 7, and Algorithm 8 detail the specific key generation, signature, and verification algorithms of the scheme, respectively.

To clarify the deterministic mapping process from the pseudo-random bitstream to matrices over  $\mathbb{F}_3$  in the implementation code, the internal instantiation mechanism of PRNG and the matrix assembly logic within this scheme are detailed below.

In the concrete implementation, PRNG is constructed based on the Chinese national standard SM3-DRNG. Given a private key seed *seed* of length  $l$  and a 16-bit random salt (e.g.,  $nonce_x, nonce_y, nonce_z, nonce_d$ ) used to derive different matrices, the algorithm concatenates these two values and inputs them into SM3-DRNG to generate an initial state, subsequently producing a pseudo-random bitstream through iterative computation. Here, the seed length  $l$  is determined by the target security level ( $l = 256$  for the 128/256-bit security levels, and  $l = 512$  for the 512-bit security level). To achieve an unbiased conversion to  $\mathbb{F}_3$  elements, the scheme adopts a word-level conversion algorithm: each output 64-bit word  $w$  is taken modulo  $3^{32}$  and expanded into ternary, extracting 32  $\mathbb{F}_3$  elements at once. Since  $2^{64}/3^{32} \approx 9955$ , the statistical bias introduced by this modulo truncation is cryptographically negligible. Subsequently, the converted elements populate the matrix in row-major order. If the generated parity-check matrices  $\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z$  are not full rank, a rejection sampling retry is performed by changing the *nonce*. For the monomial matrix  $\mathbf{D}$ , its diagonal elements are sampled independently and uniformly from  $\{1, 2\}$ , and the permutation property of the matrix is efficiently accomplished using a deterministic Fisher-Yates shuffle algorithm driven by the aforementioned bitstream.

**Analysis of Key Generation Efficiency and Retry Probability.** In the key generation algorithm, we use a 16-bit random salt *nonce* to ensure that the generated underlying matrices satisfy the full-rank condition. The expected number of executions for the **repeat...until** loops in the algorithm is analyzed below.

For the generation of random matrices over a finite field  $\mathbb{F}_q$ , the probability that a matrix of size  $r \times c$  ( $r < c$ ) is full rank is  $\prod_{i=0}^{r-1} (1 - q^{i-c})$  [11]. The probability of it not being full rank is exceedingly small, with an upper bound of approximately  $q^{-(c-r+1)}$ . In this scheme, the subcode parity-check matrices  $\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z$  all have more columns than rows. Taking  $\mathbf{H}_X$  over  $\mathbb{F}_3$  as an example, assuming its dimension is  $1600 \times 4850$ , the probability of it lacking full rank is approximately  $3^{-3251}$ , which can be considered 0 in a cryptographic sense. This means that the generation of  $\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z$  is almost certain to succeed on the first attempt.

On the other hand,  $\mathbf{H}_1$  is a square matrix of dimension  $(n - k) \times (n - k)$ . Over  $\mathbb{F}_3$ , the limit probability of a random large square matrix being invertible (i.e., full rank) is  $\prod_{j=1}^{\infty} (1 - 3^{-j}) \approx 0.5601$

[11]. Therefore, the probability that a single generation of  $\mathbf{D}$  results in an non-invertible  $\mathbf{H}_1$  is approximately 0.4399 (about 44%). Since we have allocated a 16-bit space for  $nonce_d$  (yielding a total of  $2^{16} = 65536$  possibilities), the probability of  $N$  consecutive retry failures drops exponentially. It typically takes an average of fewer than 2 attempts to find a  $\mathbf{D}$  that satisfies the condition, which does not introduce a significant performance overhead to the key generation process.

---

**Algorithm 6** UVW.KeyGen( $1^\lambda$ )

---

**Input:** Security parameter  $1^\lambda$

- 1: Determine system parameters  $\mathbf{params} = (n, k, k_1, k_2, w, l)$  according to the security parameter  $1^\lambda$ , where  $k = k_1 + k_2$
- 2: Randomly generate a master seed  $seed \xleftarrow{\$} \{0, 1\}^l$
- 3: **repeat**
- 4:   Randomly select  $nonce_x \xleftarrow{\$} \{0, 1\}^{16}$
- 5:    $\mathbf{H}_X \leftarrow \text{PRNG}(seed \parallel nonce_x) \in \mathbb{F}_3^{(n/2-k_1) \times n/2}$
- 6: **until**  $\mathbf{H}_X$  is a full-rank matrix
- 7: **repeat**
- 8:   Randomly select  $nonce_y \xleftarrow{\$} \{0, 1\}^{16}$
- 9:    $\mathbf{H}_Y \leftarrow \text{PRNG}(seed \parallel nonce_y) \in \mathbb{F}_3^{(n/2-k_1) \times n/2}$
- 10: **until**  $\mathbf{H}_Y$  is a full-rank matrix
- 11: **repeat**
- 12:   Randomly select  $nonce_z \xleftarrow{\$} \{0, 1\}^{16}$
- 13:    $\mathbf{H}_Z \leftarrow \text{PRNG}(seed \parallel nonce_z) \in \mathbb{F}_3^{(n/2-k_2) \times n/2}$
- 14: **until**  $\mathbf{H}_Z$  is a full-rank matrix
- 15: **repeat**
- 16:   Randomly select  $nonce_d \xleftarrow{\$} \{0, 1\}^{16}$
- 17:    $\mathbf{D} \leftarrow \text{PRNG}(seed \parallel nonce_d) \in \mathcal{DP}(n)$
- 18:   Compute  $\mathbf{H}_{sk} \leftarrow \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D}$
- 19:   Partition  $\mathbf{H}_{sk}$  column-wise into  $(\mathbf{H}_1 \parallel \mathbf{H}_2)$ , where  $\mathbf{H}_1$  is an  $(n - k) \times (n - k)$  square matrix
- 20: **until**  $\mathbf{H}_1$  is an invertible matrix
- 21: Transform  $(\mathbf{H}_1 \parallel \mathbf{H}_2)$  into systematic form  $(\mathbf{I}_{n-k} \parallel \mathbf{R})$  via elementary row operations

**Output:** Public parameters  $\mathbf{params}$ , public key  $\mathbf{pk} : \mathbf{R}$ ,  
private key  $\mathbf{sk} : (seed, nonce_x, nonce_y, nonce_z, nonce_d)$

---

**Algorithm 7** UVW.Sign(sk, m)

**Input:** Private key  $\text{sk} : (\text{seed}, \text{nonce}_x, \text{nonce}_y, \text{nonce}_z, \text{nonce}_d)$  and message  $\mathbf{m} \in \{0, 1\}^*$

- 1:  $\mathbf{r} \xleftarrow{\$} \{0, 1\}^\lambda$
- 2:  $\mathbf{s} \leftarrow \mathcal{H}(\mathbf{m}, \mathbf{r})$
- 3: Reconstruct matrices according to the private key:
 
$$\begin{aligned} \mathbf{H}_X &\leftarrow \text{PRNG}(\text{seed} \parallel \text{nonce}_x) \\ \mathbf{H}_Y &\leftarrow \text{PRNG}(\text{seed} \parallel \text{nonce}_y) \\ \mathbf{H}_Z &\leftarrow \text{PRNG}(\text{seed} \parallel \text{nonce}_z) \\ \mathbf{D} &\leftarrow \text{PRNG}(\text{seed} \parallel \text{nonce}_d) \end{aligned}$$
- 4: Compute parity-check matrix  $\mathbf{H}_{\text{sk}} = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D}$ , and extract its first  $n - k$  columns to obtain the square matrix  $\mathbf{H}_1$
- 5:  $(\mathbf{s}_1 \parallel \mathbf{s}_2) \leftarrow \mathbf{s} \mathbf{H}_1^\top$ , where  $\mathbf{s}_1 \in \mathbb{F}_3^{n/2-k_1}$ ,  $\mathbf{s}_2 \in \mathbb{F}_3^{n/2-k_2}$
- 6:  $\mathbf{e}_2 \leftarrow \text{Dec}_Z(\mathbf{H}_Z, \mathbf{s}_2)$
- 7:  $\mathbf{e}_1 \leftarrow \text{Dec}_{XY}(\mathbf{H}_X + \mathbf{H}_Y, \mathbf{s}_1 - \mathbf{e}_2 \mathbf{H}_Y^\top, \mathbf{e}_2)$
- 8:  $(\mathbf{e}' \parallel \mathbf{e}) \leftarrow (\mathbf{e}_1 \parallel \mathbf{e}_1 + \mathbf{e}_2)(\mathbf{D}^\top)^{-1}$ , where  $\mathbf{e}' \in \mathbb{F}_3^{n-k}$ ,  $\mathbf{e} \in \mathbb{F}_3^k$

**Output:** Signature  $\sigma = (\mathbf{e}, \mathbf{r})$

**Algorithm 8** UVW.Verify(pk, m,  $\sigma$ )

**Input:** Public key  $\text{pk} : \mathbf{R} \in \mathbb{F}_3^{(n-k) \times k}$ , message  $\mathbf{m} \in \{0, 1\}^*$ , and signature  $\sigma = (\mathbf{e}, \mathbf{r})$

- 1: **if**  $|\mathbf{e}| + |\mathcal{H}(\mathbf{m}, \mathbf{r}) - \mathbf{e} \mathbf{R}^\top| = w$  **then**
- 2:    $b \leftarrow 1$
- 3: **else**
- 4:    $b \leftarrow 0$

**Output:**  $b$

**Correctness** Let  $((\mathbf{e}' \parallel \mathbf{e}), \mathbf{r})$  be the vector returned by the signer. According to the definition of the public key matrix (i.e., transforming  $\mathbf{H}_1$  into an identity matrix via row operations is equivalent to left-multiplying by  $\mathbf{H}_1^{-1}$ , yielding  $(\mathbf{I}_{n-k} \parallel \mathbf{R}) = \mathbf{H}_1^{-1} \mathbf{H}_{\text{sk}}$ ), we have:

$$\begin{aligned} (\mathbf{I}_{n-k} \parallel \mathbf{R})(\mathbf{e}' \parallel \mathbf{e})^\top &= \mathbf{H}_1^{-1} \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D} \mathbf{D}^{-1} (\mathbf{e}_1 \parallel \mathbf{e}_1 + \mathbf{e}_2)^\top \\ &= \mathbf{H}_1^{-1} \begin{pmatrix} \mathbf{H}_X \mathbf{e}_1^\top + \mathbf{H}_Y \mathbf{e}_1^\top + \mathbf{H}_Y \mathbf{e}_2^\top \\ \mathbf{H}_Z \mathbf{e}_2^\top \end{pmatrix} \\ &= \mathbf{H}_1^{-1} \begin{pmatrix} \mathbf{s}_1^\top \\ \mathbf{s}_2^\top \end{pmatrix}. \end{aligned} \tag{1}$$

In the signature algorithm, since  $(\mathbf{s}_1 || \mathbf{s}_2) = \mathbf{s} \mathbf{H}_1^\top$ , transposing both sides yields:

$$\begin{pmatrix} \mathbf{s}_1^\top \\ \mathbf{s}_2^\top \end{pmatrix} = (\mathbf{s} \mathbf{H}_1^\top)^\top = \mathbf{H}_1 \mathbf{s}^\top. \quad (2)$$

Substituting this into equation (1), we obtain:

$$(\mathbf{I}_{n-k} || \mathbf{R})(\mathbf{e}' || \mathbf{e})^\top = \mathbf{H}_1^{-1}(\mathbf{H}_1 \mathbf{s}^\top) = \mathbf{s}^\top. \quad (3)$$

Given that  $\mathbf{s} \leftarrow \mathcal{H}(\mathbf{m}, \mathbf{r})$ , substituting and expanding this into the above equation yields:

$$\mathbf{e}' + \mathbf{e} \mathbf{R}^\top = \mathcal{H}(\mathbf{m}, \mathbf{r}), \quad i.e., \quad \mathbf{e}' = \mathcal{H}(\mathbf{m}, \mathbf{r}) - \mathbf{e} \mathbf{R}^\top, \quad (4)$$

Therefore, we only need to verify that  $|\mathbf{e}| + |\mathcal{H}(\mathbf{m}, \mathbf{r}) - \mathbf{e} \mathbf{R}^\top| = w$  holds.

## 1.6 Instantiation of Parameters and Related Functions

This scheme designs three sets of recommended parameters for the UVW signature algorithm, namely Level 1, Level 2, and Level 3. In terms of security settings, the three parameter sets correspond to classical security strengths of 160, 256, and 512 bits, respectively, and their quantum security strengths are no less than 80, 128, and 256 bits, aiming to meet the requirements against quantum attacks at different levels. The sizes of the public key  $|\mathbf{pk}|$  and signature  $|\sigma|$  in Table 1 are theoretical lower bounds calculated based on information entropy.

Table 1: Recommended Parameter Sets for the UVW Signature Scheme (Theoretical Metrics)

| Scheme           | $(q, n, k, k_1, k_2, w)$              | Public Key $ \mathbf{pk} $ | Signature $ \sigma $ |
|------------------|---------------------------------------|----------------------------|----------------------|
| UVW128 (Level 1) | (3, 9700, 4850, 3250, 1600, 8633)     | 4.44 MB                    | 977 Bytes            |
| UVW256 (Level 2) | (3, 19200, 9600, 6433, 3167, 17088)   | 17.41 MB                   | 1934 Bytes           |
| UVW512 (Level 3) | (3, 39000, 19500, 13066, 6434, 34710) | 71.85 MB                   | 3928 Bytes           |

The hash function  $\mathcal{H}$  and the pseudorandom number generator PRNG in this scheme are both instantiated using the SM3 cryptographic hash algorithm.

## 2 Design Principles

### 2.1 Hard Problems

The syndrome decoding problem is one of the most widely used problems in code-based cryptographic schemes. The security of our UVW scheme is also based on this problem.

**Problem 1** (Computational Syndrome Decoding (CSD) Problem [12]). *Given a parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$  of an  $[n, k]$  random linear code  $\mathcal{C}$ , a random vector  $\mathbf{s} \in \mathbb{F}_q^{n-k}$ , and a positive integer  $w$ . The goal is to find a vector  $\mathbf{e} \in \mathcal{E}_{n,w}$  such that  $\mathbf{e} \mathbf{H}^\top = \mathbf{s}$ .*

Problem 1 essentially asks for a solution to a system of linear equations satisfying specific conditions. This system contains  $n-k$  equations and  $n$  unknowns, hence possessing  $k$  free variables. If  $w \in [\frac{q-1}{q}(n-k), k + \frac{q-1}{q}(n-k)]$ , solving Problem 1 is relatively easy and can be efficiently accomplished by invoking the Prange algorithm (Algorithm 3) [6].

To better understand the working mechanism of Algorithm 3, we consider the last  $k$  positions of the matrix  $\mathbf{H}$  as the information set  $\mathcal{I}$ , and assume that the submatrix composed of the first  $n-k$  columns of  $\mathbf{H}$  is invertible. Given the matrix  $\mathbf{H} = (\mathbf{A}||\mathbf{B})$  and the vector  $\mathbf{s}$ , the goal is to find a vector  $\mathbf{e} = (\mathbf{e}'||\mathbf{e}'')$  (where  $\mathbf{e}' \in \mathbb{F}_q^{n-k}, \mathbf{e}'' \in \mathbb{F}_q^k$ ) such that  $\mathbf{e}\mathbf{H}^\top = \mathbf{s}$ . Obviously,  $\mathbf{e}' = (\mathbf{s} - \mathbf{e}''\mathbf{B}^\top)(\mathbf{A}^{-1})^\top$ , which implies that once  $\mathbf{e}''$  is determined,  $\mathbf{e}$  is also uniquely determined. Since the operations are defined over the finite field  $\mathbb{F}_q$ , the probability of each component in  $\mathbf{e}'$  being 0 is equal to  $\frac{1}{q}$ ; thus, the expected weight of  $\mathbf{e}'$  is  $\mathbb{E}(|\mathbf{e}'|) = \frac{q-1}{q}(n-k)$ . Following the previous analysis, if we wish to minimize the weight of  $\mathbf{e}$ , we can set all components of the vector  $\mathbf{e}''$  to 0, yielding  $\mathbb{E}(|\mathbf{e}|) = \frac{q-1}{q}(n-k)$ . If we wish to maximize the weight of  $\mathbf{e}$ , we can set all components of the vector  $\mathbf{e}''$  to non-zero values, yielding  $\mathbb{E}(|\mathbf{e}|) = k + \frac{q-1}{q}(n-k)$ . This means that for any integer  $w \in [\frac{q-1}{q}(n-k), k + \frac{q-1}{q}(n-k)]$ , by appropriately choosing the weight of  $\mathbf{e}''$ , we can obtain an  $\mathbf{e} \in \mathcal{E}_{n,w}$  such that  $\mathbf{e}\mathbf{H}^\top = \mathbf{s}$  with an overwhelmingly high probability. This probability is expressed as follows:

$$\begin{aligned} \mathbb{P}(\mathbf{e} \leftarrow \text{Prange}(\mathbf{H}, \mathbf{s}, w), |\mathbf{e}| = w, \mathbf{e}\mathbf{H}^\top = \mathbf{s} \mid w \in [\frac{q-1}{q}(n-k), k + \frac{q-1}{q}(n-k)]) \\ = \frac{\binom{n-k}{\frac{q-1}{q}(n-k)}(q-1)^{\frac{q-1}{q}(n-k)}}{q^{n-k}}. \end{aligned} \quad (5)$$

For values outside the interval  $[\frac{q-1}{q}(n-k), k + \frac{q-1}{q}(n-k)]$ , the problem becomes hard to solve, requiring an exponential number of repeated invocations of the Prange algorithm. Figure 2 provides a brief schematic illustration of the hardness of Problem 1 when  $w$  falls into different intervals.

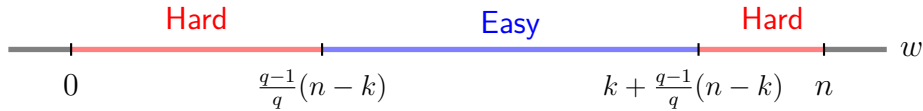


Figure 2: Hardness of Problem 1 across different intervals of  $w$

Since an adversary can query the hash oracle on multiple messages to obtain a polynomial number of syndromes, we present a variant of Problem 1, which is equally hard and better suited for proving the security of the scheme.

**Problem 2** (Decoding One Out of Many (DOOM) [13]). *Given a parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$  of an  $[n, k]$  random linear code  $\mathcal{C}$ ,  $N$  random vectors  $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N \in \mathbb{F}_q^{n-k}$ , and a positive integer  $w$ . The goal is to find a vector  $\mathbf{e} \in \mathcal{E}_{n,w}$  and an integer  $i \in [N]$  such that  $\mathbf{e}\mathbf{H}^\top = \mathbf{s}_i$ .*

## 2.2 Main Design Ideas and Algebraic Structure

### 2.2.1 The UVW Trapdoor

The trapdoor of the UVW scheme is precisely the generalized  $(U + V, U + W)$  code utilized in the scheme. It can expand the upper bound of the effective decoding interval for random codes from  $k + \frac{q-1}{q}(n-k)$  to  $\frac{q-1}{q}n + \frac{2k_1}{q}$ . In our scheme, we use large-weight vectors as signatures, i.e.,  $w > k + \frac{q-1}{q}(n-k)$ . Next, we present an analysis of how the UVW trapdoor expands the upper bound of the effective decoding interval.

Choose a generalized  $[n, k]$   $(U + V, U + W)$  code  $\mathcal{C}$ . Let

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D}$$

be the parity-check matrix of code  $\mathcal{C}$ , where  $\mathbf{H}_X, \mathbf{H}_Y \in \mathbb{F}_q^{(n/2-k_1) \times n/2}$ ,  $\mathbf{H}_Z \in \mathbb{F}_q^{(n/2-k_2) \times n/2}$ ,  $k = k_1 + k_2$ . Let  $\mathbf{s} = (\mathbf{s}_1 || \mathbf{s}_2)$ , where  $\mathbf{s}_1 \in \mathbb{F}_q^{n/2-k_1}$ ,  $\mathbf{s}_2 \in \mathbb{F}_q^{n/2-k_2}$ . The goal is to find a vector  $\mathbf{e} \in \mathcal{E}_{n,w}$  such that

$$\mathbf{e} \mathbf{H}^\top = \mathbf{s}. \quad (6)$$

We split the vector  $\mathbf{e}$  into  $\mathbf{e} = (\mathbf{e}_1 || \mathbf{e}_1 + \mathbf{e}_2)$ , then solving (6) is equivalent to solving the following two equations:

$$\begin{aligned} \mathbf{e}_2 \mathbf{H}_Z^\top &= \mathbf{s}_2, \\ \mathbf{e}_1 \mathbf{H}_X^\top + \mathbf{e}_1 \mathbf{H}_Y^\top + \mathbf{e}_2 \mathbf{H}_Y^\top &= \mathbf{s}_1. \end{aligned} \quad (7)$$

Since  $\mathbf{D}$  is a monomial matrix (the product of an invertible diagonal matrix and a permutation matrix), multiplying it with any vector does not change the weight of the vector, so we ignore matrix  $\mathbf{D}$  here. When  $q = 2$ , finding a small-weight vector and finding a large-weight vector are equivalent. However, when  $q \geq 3$ , the situation is different. Here we only consider the case where  $q \geq 3$ . From Equation (7), it can be seen that we first need to find vector  $\mathbf{e}_2$ , and then find vector  $\mathbf{e}_1$ . We observe that regardless of how the weight of vector  $\mathbf{e}_2$  is distributed, the total weight of vector  $\mathbf{e}$  is completely determined by vector  $\mathbf{e}_1$ . Recall the working principle of the Prange algorithm. First, select an information set  $\mathcal{I}$  of size  $k_1$ , and determine the values of vector  $\mathbf{e}_1$  at these  $k_1$  positions, then use the equation  $\mathbf{e}_1 \mathbf{H}_X^\top + \mathbf{e}_1 \mathbf{H}_Y^\top + \mathbf{e}_2 \mathbf{H}_Y^\top = \mathbf{s}_1$  to calculate the values for the remaining  $\frac{n}{2} - k_1$  positions of vector  $\mathbf{e}_1$ . To maximize the weight of vector  $\mathbf{e}$ , for any  $i \in \mathcal{I}$ , we choose a value that makes both  $\mathbf{e}_1(i)$  and  $\mathbf{e}_1(i) + \mathbf{e}_2(i)$  non-zero. At this point, the expectation of the weight of vector  $\mathbf{e}$  is:

$$\begin{aligned} \mathbb{E}(|\mathbf{e}|) &= 2k_1 + \frac{q-1}{q}(n-2k_1) \\ &= \frac{q-1}{q}n + \frac{2k_1}{q}. \end{aligned}$$

To ensure that the upper bound of the effective decoding interval increases, the parameter selection must satisfy the following inequality:

$$\frac{q-1}{q}n + \frac{2k_1}{q} > k + \frac{q-1}{q}(n-k) \Leftrightarrow k_1 > k_2.$$

### 2.2.2 Analysis of the Evolution and Degeneration Relationship with the Wave Scheme

In Section 1.2, we provided the definition of the standard  $(U, U + V)$  code. To overcome the weakness of the standard  $(U, U + V)$  code (as described in Definition 2), which is vulnerable to structural recovery attacks due to the presence of zero matrix blocks, the Wave signature scheme [5] proposed the concept of the generalized  $(U, U + V)$  code. By introducing secret vector multipliers  $b, c$  and  $d = 1 + bc$ , it algebraically couples the parity-check matrices  $\mathbf{H}_U$  and  $\mathbf{H}_V$  of the subcodes, constructing the following classical trapdoor skeleton:

$$\mathbf{H}_{Wave} = \begin{pmatrix} d \cdot \mathbf{H}_U & -b \cdot \mathbf{H}_U \\ -c \cdot \mathbf{H}_V & \mathbf{H}_V \end{pmatrix}$$

To clarify the design origin of our scheme and its universality in algebraic structure, the following content in this section will detail the intrinsic connection between the UVW trapdoor structure and the classical Wave scheme. The analysis shows that structurally, the Wave scheme can be viewed as a degenerate special case of the UVW scheme under specific parameter constraints.

The parity-check matrix  $\mathbf{H}_{sk}$  of the UVW scheme is constructed as follows:

$$\mathbf{H}_{sk} = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D}$$

where  $\mathbf{D}$  is a monomial matrix containing a diagonal matrix and a permutation matrix.

When the subcode matrices of the UVW scheme satisfy the following five mapping conditions, its trapdoor structure will be entirely equivalent to the Wave scheme:

1. Let  $\mathbf{D} = \begin{pmatrix} \text{diag}(c) & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n/2} \end{pmatrix}$ , where  $\text{diag}(c)$  denotes a diagonal matrix whose main diagonal elements are the elements of the secret vector  $c$ . Under this condition,  $\mathbf{H}_{sk}$  is expanded via matrix multiplication to  $\begin{pmatrix} c \cdot \mathbf{H}_X & \mathbf{H}_Y \\ -c \cdot \mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix}$ .

2. Bottom-right alignment: Let  $\mathbf{H}_Z = \mathbf{H}_V$ .

3. Bottom-left alignment: We have  $-c \cdot \mathbf{H}_Z = -c \cdot \mathbf{H}_V$ . Given that condition 1 holds, this equation holds naturally.

4. Top-right alignment: Let  $\mathbf{H}_Y = -b \cdot \mathbf{H}_U$ .

5. Top-left alignment: Over the field  $\mathbb{F}_3$ , the inverse of a non-zero element equals itself (i.e.,  $c^{-1} = c$ ), and  $c^2 = 1$ . For  $c \cdot \mathbf{H}_X = d \cdot \mathbf{H}_U$  to hold, substituting  $d = 1 + bc$  and multiplying both sides by  $c$  yields:  $\mathbf{H}_X = (c + b) \cdot \mathbf{H}_U$ .

The above derivation proves that, while preserving the security properties of the Wave scheme, the UVW scheme greatly expands the key space by decoupling the generation logic of  $\mathbf{H}_X, \mathbf{H}_Y, \mathbf{H}_Z$  from pure vector coupling.

## 2.3 Rationale for Parameter Selection and Failure Probability Analysis

In selecting system parameters, this scheme primarily balances the reliability of the algorithm and its resistance to side-channel attacks. Although  $k_1/k \approx 0.696$  is the theoretical security balance point, in engineering practice, this ratio causes the weight distribution center of the candidate signatures to be significantly higher than the target value  $w$ , thereby leading to an extremely high rejection sampling rate. Figure 9 (detailed in Section 3.3 below) illustrates the trend of quantum security varying with the ratio  $k_1/k$  under these specifications. The analysis shows that as  $k_1/k$  decreases, the complexity of forgery attacks remains largely stable, but the complexity of key recovery attacks increases significantly, providing greater security redundancy for the system.

### 2.3.1 Failure Probability and Distribution Matching

To ensure the efficiency of signature generation and achieve the decoupling of computation time from secret information, this scheme selects  $k_1/k \approx 0.67$  as the working point. As shown in the following theorem, this choice maximizes the success rate of a single sampling.

**Theorem 2** (Bound on Signature Sampling Success Rate). *Let  $\mathbf{e} = (\mathbf{e}_1 \parallel \mathbf{e}_1 + \mathbf{e}_2)$  be the candidate signature vector generated in a single iteration of the UVW signature scheme. Given the parameters  $(q, n, k_1, w)$ , if the output of the underlying random linear system follows a uniform distribution over the finite field  $\mathbb{F}_q$ , then the probability  $\mathbb{P}_{succ}$  that a single sampling satisfies the target Hamming weight  $wt(\mathbf{e}) = w$  is:*

$$\mathbb{P}_{succ} = \frac{\binom{n-2k_1}{w-2k_1} (q-1)^{w-2k_1}}{q^{n-2k_1}}$$

Furthermore, if a hard truncation of  $N$  retries is implemented in engineering practice, the upper bound of the probability  $\mathbb{P}_{fail}(N)$  that the algorithm fails to generate a signature due to timeout is:

$$\mathbb{P}_{fail}(N) = \left( 1 - \frac{\binom{n-2k_1}{w-2k_1} (q-1)^{w-2k_1}}{q^{n-2k_1}} \right)^N$$

**Proof.** During the execution of the signature generation algorithm, the generation of the candidate signature vector  $\mathbf{e}$  depends on the selection of the information set  $\mathcal{I}$  and the execution of the Prange algorithm. By construction, the size of the information set is  $k_1$ . To maximize the weight of the vector, when setting the free variables, the algorithm ensures that the components at these positions and their associated positions are all non-zero. Therefore, the number of non-zero coordinates in vector  $\mathbf{e}$  completely determined by the information set is  $2k_1$ , meaning the deterministic weight contributed by this part is  $2k_1$ . The total length of vector  $\mathbf{e}$  is  $n$ . After excluding the  $2k_1$  deterministic coordinates, the values of the remaining  $n - 2k_1$  coordinates are entirely determined by solving a system of linear equations using the syndrome and the remaining random matrix part. Under the random oracle model and the pseudorandom matrix assumption, these  $n - 2k_1$  coordinates can be considered independent and uniformly distributed over the finite field  $\mathbb{F}_q$ .

Define a random variable  $X$  as the number of non-zero components among these  $n - 2k_1$  free coordinates. For a uniform distribution over  $\mathbb{F}_q$ , the probability of each coordinate being zero is  $\frac{1}{q}$ ,

and the probability of being non-zero is  $\frac{q-1}{q}$ . Thus,  $X$  strictly follows a binomial distribution:

$$X \sim B\left(n - 2k_1, \frac{q-1}{q}\right)$$

The algorithm requires the total weight of the output signature to be strictly equal to  $w$ . Since the deterministic part has already contributed a weight of  $2k_1$ , the free coordinate part must contribute exactly a weight of  $w - 2k_1$ . The probability of a successful single sampling is precisely the probability of the event  $\{X = w - 2k_1\}$  occurring:

$$\mathbb{P}_{succ} = \mathbb{P}(X = w - 2k_1) = \binom{n - 2k_1}{w - 2k_1} \left(\frac{q-1}{q}\right)^{w-2k_1} \left(\frac{1}{q}\right)^{(n-2k_1)-(w-2k_1)}$$

Simplifying the above equation yields:

$$\mathbb{P}_{succ} = \frac{\binom{n-2k_1}{w-2k_1} (q-1)^{w-2k_1}}{q^{n-2k_1}}$$

Since the generation process of each candidate vector (selecting a new random salt or perturbation) is statistically independent, the probability of the event that  $N$  consecutive iterations fail to hit the target weight is the single failure probability raised to the power of  $N$ . Therefore, with a hard truncation at a maximum of  $N$  iterations, the probability that the algorithm fails to output a valid signature is:

$$\mathbb{P}_{fail}(N) = (1 - \mathbb{P}_{succ})^N = \left(1 - \frac{\binom{n-2k_1}{w-2k_1} (q-1)^{w-2k_1}}{q^{n-2k_1}}\right)^N$$

□

### 2.3.2 Conclusion and Connection to Security

Based on the derivation of Theorem 2, the weight of the random part of the candidate signature strictly follows a binomial distribution. When  $k_1/k \approx 0.696$ , the distribution center  $\mu$  is significantly higher than the target value  $w$ , resulting in an extremely low  $\mathbb{P}_{succ}$ . This not only drastically reduces signature efficiency but also tends to introduce timing side-channel risks due to drastic fluctuations in the number of sampling iterations. Conversely, when  $k_1/k \approx 0.67$ , theoretical calculations indicate that the distribution center  $\mu$  is highly aligned with the target weight  $w$  (with an error of  $< 1$ ). Taking Level 1 as an example ( $n = 9700, k_1 = 3250$ ), the target weight for the random part is 2133, while the theoretical expectation of the binomial distribution is 2133.33, achieving a high single sampling success rate of approximately 1.5%.

By selecting  $k_1/k \approx 0.67$ , this scheme introduces a hard truncation mechanism matching the security level in its engineering implementation. By setting the truncation threshold  $N$ , it ensures that the failure probability of rejection sampling (i.e., the correctness error of the scheme)  $\mathbb{P}_{fail} = (1 - \mathbb{P}_{succ})^N$  is strictly lower than the stated security bound of the scheme (for instance, setting  $N = 6000$  for Level 1 to satisfy  $\mathbb{P}_{fail} < 2^{-128}$ ; and setting  $N = 12000$  for Level 2 to satisfy

$\mathbb{P}_{fail} < 2^{-256}$ ). Since the average number of sampling iterations required is only about  $1/\mathbb{P}_{succ} \approx 67$ , increasing  $N$  merely seals off extremely low-probability theoretical attack paths at the mathematical level, without causing any substantial impact on the average signature speed. This design guarantees the efficient generation of signatures while achieving a complete statistical decoupling of signature timing from secret information, thereby providing robust resistance against timing side-channel attacks.

While ensuring the aforementioned engineering properties, the remaining free parameter space of this scheme is still sufficient to provide security redundancy exceeding the claimed levels. For a quantitative evaluation of the complexity and specific security bits of this selection scheme against existing cryptanalytic methods such as Information Set Decoding attacks and Wagner’s algorithm, please refer to Section 3.3, “Concrete Security”, of this paper.

### 3 Security Statement and Analysis

#### 3.1 Algorithm Security Statement

The proposed UVW digital signature algorithm satisfies Existential Unforgeability under Adaptive Chosen-Message Attacks (EUF-CMA) in the Random Oracle Model. Targeting the three declared classical security strengths of 128, 256, and 512 bits, the actual complexity against both classical and quantum computing attacks for each instance strictly meets the standards, with quantum security strengths not lower than 80, 128, and 256 bits, respectively. The design is based on solid hard problems in coding theory (the CSD and the DOOM problem). No backdoors are reserved, and it contains no components known to be vulnerable to quantum attacks, ensuring sufficient security redundancy.

#### 3.2 Theoretical Security

##### 3.2.1 Definitions of Digital Signature and Security

The definitions of the signature scheme and the security model are provided below.

**Definition 5** (Digital Signature [14]). *A signature scheme  $\mathcal{S}$  consists of three Probabilistic Polynomial Time algorithms: KeyGen, Sign, and Verify. The details are as follows:*

- **KeyGen**( $1^\lambda$ ): *Takes the security parameter  $\lambda$  as input and outputs a public-private key pair  $(pk, sk)$ .*
- **Sign**( $sk, \mathbf{m}$ ): *A Probabilistic Polynomial Time (PPT) algorithm that takes the private key  $sk$  and a message  $\mathbf{m} \in \{0, 1\}^*$  as input and outputs a signature  $\sigma = \text{Sign}(sk, \mathbf{m})$ .*
- **Verify**( $pk, \mathbf{m}, \sigma$ ): *Takes the public key  $pk$  and a message-signature pair  $(\mathbf{m}, \sigma)$  as input and outputs a bit  $b \in \{0, 1\}$ .*

If  $\sigma$  is a valid signature for message  $\mathbf{m}$  under public key  $\mathbf{pk}$ , the algorithm outputs “1” (accept); otherwise, it outputs “0” (reject). Specifically, for any key pair  $(\mathbf{pk}, \mathbf{sk}) \leftarrow \text{KeyGen}(1^\lambda)$  and any message  $\mathbf{m}$ , if  $\sigma$  is the output of  $\text{Sign}(\mathbf{sk}, \mathbf{m})$ , then  $(\mathbf{m}, \sigma)$  is always accepted by the verification algorithm, i.e.,

$$\text{Verify}(\mathbf{pk}, \mathbf{m}, \sigma) = 1.$$

Our signature scheme follows the “hash-and-sign” paradigm [15], which involves hashing the message first and subsequently signing the resulting hash value. This paradigm has been widely applied in trapdoor-based post-quantum cryptosystems (such as the lattice-based GPV framework proposed by Gentry et al. [16]). Under this framework, the signing process is essentially transformed into finding a preimage of the target hash value subject to a specific statistical distribution. This not only effectively mitigates information leakage attacks but also provides a strong provable security guarantee for the scheme.

Typically, the hash function is modeled as a random oracle. EUF-CMA is one of the most widely used security models for signature schemes. In this attack model, an adversary is allowed to query the hash oracle up to  $Q_{\mathcal{H}}$  times and the signing oracle up to  $Q_{\Sigma} < Q_{\mathcal{H}}$  times, with the objective of outputting at least one valid message-signature pair that passes verification. The formal definition of the EUF-CMA security for a signature scheme is given as follows:

**Definition 6.** Let  $\mathcal{S}$  be a signature scheme. An adversary  $\mathcal{A}$  is considered a  $(\tau, \epsilon, Q_{\mathcal{H}}, Q_{\Sigma})$ -adversary for  $\mathcal{S}$  in the EUF-CMA model if, after querying the hash oracle  $\mathcal{H}$  for  $Q_{\mathcal{H}}$  times, the signature oracle for  $Q_{\Sigma} < Q_{\mathcal{H}}$  times, and running for time  $\tau$ , it can output a valid forgery with a probability of at least  $\epsilon$ . We define the EUF-CMA success probability of scheme  $\mathcal{S}$  as:

$$\text{Succ}_{\mathcal{S}}^{\text{EUF-CMA}}(\tau, Q_{\mathcal{H}}, Q_{\Sigma}) := \max_{\mathcal{A}}(\epsilon \mid \mathcal{A} \text{ is a } (\tau, \epsilon, Q_{\mathcal{H}}, Q_{\Sigma})\text{-adversary}).$$

The signature scheme  $\mathcal{S}$  is  $(\tau, Q_{\mathcal{H}}, Q_{\Sigma})$ -secure in the EUF-CMA model if the above probability is a negligible function of the security parameter  $\lambda$ .

### 3.2.2 Security Assumptions

As previously discussed, the hardness of the DOOM problem is defined over random codes. However, our signature scheme is constructed using generalized  $(U + V, U + W)$  codes. Therefore, we must consider the hardness of the Generalized  $(U + V, U + W)$  Code Distinguishability Problem (GDP).

Let  $\mathcal{H}_{\mathbf{pk}}$  be the set of all public key matrices in the UVW signature scheme. A GDP-distinguisher  $\mathcal{D}$  is a PPT algorithm. We define its advantage as follows:

$$\text{Adv}^{\text{GDP}}(\mathcal{D}) = \left| \mathbb{P}[\mathbf{H} \xleftarrow{\$} \mathcal{H}_{\mathbf{pk}} \text{ code } [n, k] : \mathcal{D}(\mathbf{H}) = 1] - \mathbb{P}[\mathbf{H} \xleftarrow{\$} \text{ random code } [n, k] : \mathcal{D}(\mathbf{H}) = 1] \right|. \quad (8)$$

If this advantage is negligible for any distinguisher  $\mathcal{D}$ , then generalized  $(U + V, U + W)$  codes are indistinguishable from random linear codes.

**Definition 7.** Let  $\mathcal{H}_{\text{pk}}$  be the set of all public key matrices in the UVW signature scheme. The Generalized  $(U + V, U + W)$  Code Distinguishability Problem is  $(\tau, \epsilon)$ -secure if, for any distinguisher  $\mathcal{D}$  with a running time of at most  $\tau$ , we have:

$$\text{Adv}^{\text{GDP}}(\mathcal{D}) \leq \epsilon.$$

### 3.2.3 Domain Sampling with Uniform Output

In this section, we prove that the statistical distance between the probability distribution of syndromes generated by public key matrices and the distribution of syndromes chosen uniformly at random is negligible.

To this end, we first introduce three important propositions and lemmas.

**Proposition 3.** [5] Let  $(\mathbf{u} \parallel \mathbf{v})$  be a random variable following the uniform distribution  $\mathcal{U}_w$ , where  $\mathbf{u}, \mathbf{v} \in \mathbb{F}_3^{n/2}$ . The probability distribution of the weight of vector  $\mathbf{v} - \mathbf{u}$  is as follows:

$$\mathbb{P}(|\mathbf{v} - \mathbf{u}| = t) = \frac{1}{\binom{n}{w} 2^w} \sum_{\substack{i = \max\{0, 2t - w\} \\ w + i \equiv 0 \pmod{2}}}^t \binom{n/2}{i} \binom{n/2 - i}{t - i} \binom{n/2 - t}{(w + i)/2 - t} 2^{\frac{w + 3i}{2}}, \text{ with } t \in [0, \frac{n}{2}]. \quad (9)$$

**Lemma 1.** [5] Let  $\mathcal{F}$  be a finite family of functions mapping from a finite set  $D$  to a finite set  $E$ . Let  $\theta$  be the deviation of the collision probability, i.e.,

$$\mathbb{P}(f(e) = f(e')) = \frac{1}{|E|} (1 + \theta),$$

where  $f \xleftarrow{\$} \mathcal{F}$  and  $e, e' \xleftarrow{\$} D$ . Let  $\mathcal{U}$  be the uniform distribution over set  $E$ , and  $\mathcal{D}_f$  be the distribution of  $f(e)$  where  $e \xleftarrow{\$} D$ . Then:

$$\mathbb{E}_f(\rho(\mathcal{D}_f, \mathcal{U})) \leq \frac{1}{2} \sqrt{\theta}.$$

Interested readers may refer to [5] for more details regarding Lemma 1.

**Lemma 2.** [5] Let  $\mathbf{H} \in \mathbb{F}_3^{r \times n}$  ( $r \leq n$ ) be a random full-rank matrix and  $\mathbf{s} \in \mathbb{F}_3^r$  be a random vector. For any non-zero vector  $\mathbf{x} \in \mathbb{F}_3^n$ , we have:

$$\mathbb{P}(\mathbf{H}\mathbf{x}^\top = \mathbf{s}^\top) = \frac{1}{3^r}.$$

With the above propositions and lemmas, we will now prove that the statistical distance between the probability distribution of syndromes generated by public key matrices and that of syndromes chosen uniformly at random is negligible.

**Theorem 4.** Let  $\mathbf{H}$  be an  $(n - k) \times n$  matrix over the finite field  $\mathbb{F}_3$ , and let  $\mathcal{D}_{\mathbf{H}, w}$  be the probability distribution of the syndrome  $\mathbf{e}\mathbf{H}^\top$ , where  $\mathbf{e} \xleftarrow{\$} \mathcal{E}_{n, w}$ . Let  $\mathcal{U}$  be the uniform distribution over the syndrome space  $\mathbb{F}_3^{n-k}$ . Then,

$$\mathbb{E}_{\mathbf{H}_{\text{pk}}}(\rho(\mathcal{D}_{\mathbf{H}_{\text{pk}}, w}, \mathcal{U})) \leq \frac{1}{2} \sqrt{\theta},$$

where

$$\theta \leq \frac{3^{n-k}}{\binom{n}{w} 2^w} + \sum_{t=0}^{n/2} \frac{3^{n/2-k_2}}{\binom{n/2}{t} 2^t \binom{n}{w}^2 2^{2w}} \left( \sum_{\substack{i = \max\{0, 2t-w\} \\ w+i \equiv 0 \pmod{2}}}^t \binom{n/2}{i} \binom{n/2-i}{t-i} \binom{n/2-t}{(w+i)/2-t} 2^{\frac{w+3i}{2}} \right)^2.$$

**Proof.** Let  $\mathbf{H}_{\text{pk}} \in \mathbb{F}_3^{(n-k) \times n}$  be an arbitrary public key matrix in the set  $\mathcal{H}_{\text{pk}}$ , and let  $\mathbf{x}$  and  $\mathbf{y}$  be chosen uniformly at random from the set  $\mathcal{E}_{n,w}$ . Then,

$$\mathbb{P}(\mathbf{H}_{\text{pk}} \mathbf{x}^\top = \mathbf{H}_{\text{pk}} \mathbf{y}^\top) = \frac{1}{3^{n-k}} (1 + \theta).$$

Recall the definition of the public key matrix  $\mathbf{H}_{\text{pk}}$ :

$$\mathbf{H}_{\text{pk}} = \mathbf{S} \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{D},$$

where  $\mathbf{H}_X, \mathbf{H}_Y \in \mathbb{F}_3^{(n/2-k_1) \times n/2}$  and  $\mathbf{H}_Z \in \mathbb{F}_3^{(n/2-k_2) \times n/2}$  are three random full-rank matrices,  $\mathbf{S} \in \mathbb{F}_3^{(n-k) \times (n-k)}$  is a random non-singular matrix, and  $\mathbf{D} \in \mathbb{F}_3^{n \times n}$  is a random monomial matrix (a permutation matrix multiplied by an invertible diagonal matrix).

Given the public key matrix  $\mathbf{H}_{\text{pk}}$ , we choose two vectors  $\mathbf{x}$  and  $\mathbf{y}$  uniformly at random from the set  $\mathcal{E}_{n,w}$ . Since  $\mathbf{S}$  is a non-singular matrix and  $\mathbf{D}$  is a monomial matrix, the probability of the event  $A : \mathbf{H}_{\text{pk}} \mathbf{x}^\top = \mathbf{H}_{\text{pk}} \mathbf{y}^\top$  occurring is equal to the probability of the event

$$B : \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{x}^\top = \begin{pmatrix} \mathbf{H}_X & \mathbf{H}_Y \\ -\mathbf{H}_Z & \mathbf{H}_Z \end{pmatrix} \mathbf{y}^\top \quad (10)$$

occurring.

To better calculate the probability of event  $B$ , we partition the sample space into two disjoint events based on  $\mathbf{x} - \mathbf{y}$ :  $\mathcal{E}_1 : \mathbf{x} - \mathbf{y} = \mathbf{0}$  and  $\mathcal{E}_2 : \mathbf{x} - \mathbf{y} \neq \mathbf{0}$ . Then,

$$\mathbb{P}(B) = \mathbb{P}(B|\mathcal{E}_1)\mathbb{P}(\mathcal{E}_1) + \mathbb{P}(B|\mathcal{E}_2)\mathbb{P}(\mathcal{E}_2), \quad (11)$$

and

$$\mathbb{P}(\mathcal{E}_1) = \frac{1}{\binom{n}{w} 2^w}, \text{ and } \mathbb{P}(\mathcal{E}_2) = 1 - \frac{1}{\binom{n}{w} 2^w}. \quad (12)$$

Obviously, when  $\mathbf{x} - \mathbf{y} = \mathbf{0}$ , event  $B$  always occurs. Below, we focus on analyzing the probability of event  $B$  when  $\mathbf{x} - \mathbf{y} \neq \mathbf{0}$ .

Given that  $\mathbf{x} - \mathbf{y} \neq \mathbf{0}$ , we split the vectors  $\mathbf{x}$  and  $\mathbf{y}$  into equal-length halves:  $\mathbf{x} = (\mathbf{x}_1 || \mathbf{x}_2)$  and  $\mathbf{y} = (\mathbf{y}_1 || \mathbf{y}_2)$ . Consequently, the vectors  $\mathbf{x}_1 - \mathbf{y}_1$  and  $\mathbf{x}_2 - \mathbf{y}_2$  cannot be simultaneously  $\mathbf{0}$ . Since  $\mathbf{H}_X$  and  $\mathbf{H}_Y$  are random full-rank matrices, according to Lemma 2, we have:

$$\begin{aligned} \mathbb{P}(B|\mathcal{E}_2) &= \mathbb{P}((\mathbf{x}_1 - \mathbf{y}_1)\mathbf{H}_X^\top + (\mathbf{x}_2 - \mathbf{y}_2)\mathbf{H}_Y^\top = \mathbf{0}, ((\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2))\mathbf{H}_Z^\top = \mathbf{0} | \mathcal{E}_2) \\ &= \frac{1}{3^{n/2-k_1}} \mathbb{P}(((\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2))\mathbf{H}_Z^\top = \mathbf{0} | \mathcal{E}_2). \end{aligned} \quad (13)$$

To compute  $\mathbb{P}(((\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2))\mathbf{H}_Z^\top = \mathbf{0} \mid \mathcal{E}_2)$ , we consider two disjoint events:  $\mathcal{E}_{2,1} : (\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2) = \mathbf{0}$  and  $\mathcal{E}_{2,2} : (\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2) \neq \mathbf{0}$ . Since  $\mathbf{H}_Z$  is a random full-rank matrix, according to Lemma 2, we obtain:

$$\begin{aligned} \mathbb{P}(((\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2))\mathbf{H}_Z^\top = \mathbf{0} \mid \mathcal{E}_2) &= \mathbb{P}(\mathcal{E}_{2,1} \mid \mathcal{E}_2) + \frac{1}{3^{n/2-k_2}} \mathbb{P}(\mathcal{E}_{2,2} \mid \mathcal{E}_2) \\ &\leq \mathbb{P}(\mathcal{E}_{2,1} \mid \mathcal{E}_2) + \frac{1}{3^{n/2-k_2}}. \end{aligned} \quad (14)$$

When  $\mathbf{x} - \mathbf{y} = \mathbf{0}$ , it strictly follows that  $(\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2) = \mathbf{0}$ , thus we have:

$$\mathbb{P}(\mathcal{E}_{2,1} \mid \mathcal{E}_1) = 1, \quad (15)$$

Furthermore,

$$\mathbb{P}(\mathcal{E}_{2,1} \mid \mathcal{E}_2) \leq \mathbb{P}(\mathcal{E}_{2,1}). \quad (16)$$

Given  $(\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2) = \mathbf{0}$ , we can deduce that  $\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{y}_1 - \mathbf{y}_2$ . Let  $\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e}$  and  $|\mathbf{x}_1 - \mathbf{x}_2| = w_e$ . According to Proposition 3, we obtain:

$$\begin{aligned} \mathbb{P}(\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e}) &= \mathbb{P}(\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e} \mid |\mathbf{x}_1 - \mathbf{x}_2| = w_e) \mathbb{P}(|\mathbf{x}_1 - \mathbf{x}_2| = w_e) \\ &= \frac{1}{\binom{n/2}{w_e} 2^{w_e}} \frac{1}{\binom{n}{w} 2^w} \sum_{\substack{i = \max\{0, 2t - w\} \\ w + i \equiv 0 \pmod{2}}}^{w_e} \binom{n/2}{i} \binom{n/2 - i}{w_e - i} \binom{n/2 - w_e}{(w + i)/2 - w_e} 2^{\frac{w+3i}{2}}. \end{aligned} \quad (17)$$

Therefore, we have:

$$\begin{aligned} \mathbb{P}(\mathcal{E}_{2,1}) &= \mathbb{P}(\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{y}_1 - \mathbf{y}_2) \\ &= \sum_{t=0}^{n/2} \sum_{\mathbf{e} \in \mathbb{F}_3^{n/2} : |\mathbf{e}|=t} \mathbb{P}(\mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e})^2 \\ &= \sum_{t=0}^{n/2} \binom{n/2}{t} 2^t \left( \frac{1}{\binom{n/2}{t} 2^t} \frac{1}{\binom{n}{w} 2^w} \sum_{\substack{i = \max\{0, 2t - w\} \\ w + i \equiv 0 \pmod{2}}}^t \binom{n/2}{i} \binom{n/2 - i}{t - i} \binom{n/2 - t}{(w + i)/2 - t} 2^{\frac{w+3i}{2}} \right)^2 \quad (18) \\ &= \sum_{t=0}^{n/2} \frac{1}{\binom{n/2}{t} 2^t \binom{n}{w}^2 2^{2w}} \left( \sum_{\substack{i = \max\{0, 2t - w\} \\ w + i \equiv 0 \pmod{2}}}^t \binom{n/2}{i} \binom{n/2 - i}{t - i} \binom{n/2 - t}{(w + i)/2 - t} 2^{\frac{w+3i}{2}} \right)^2. \end{aligned}$$

Based on Equations (11), (12), (13), (14), (16), and (18), we obtain:

$$\begin{aligned}
\mathbb{P}(B) &= \mathbb{P}(B|\mathcal{E}_1)\mathbb{P}(\mathcal{E}_1) + \mathbb{P}(B|\mathcal{E}_2)\mathbb{P}(\mathcal{E}_2) \\
&= \frac{1}{\binom{n}{w}2^w} + (1 - \frac{1}{\binom{n}{w}2^w})\mathbb{P}(B|\mathcal{E}_2) \\
&\leq \frac{1}{\binom{n}{w}2^w} + \frac{1}{3^{n/2-k_1}}\mathbb{P}(((\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2))\mathbf{H}_Z^\top = \mathbf{0} \mid \mathcal{E}_2) \\
&\leq \frac{1}{\binom{n}{w}2^w} + \frac{1}{3^{n/2-k_1}}\left(\mathbb{P}(\mathcal{E}_{2,1} \mid \mathcal{E}_2) + \frac{1}{3^{n/2-k_2}}\right) \\
&\leq \frac{1}{\binom{n}{w}2^w} + \frac{1}{3^{n/2-k_1}}\left(\mathbb{P}(\mathcal{E}_{2,1}) + \frac{1}{3^{n/2-k_2}}\right) \\
&= \frac{1}{3^{n-k}}\left(1 + \frac{3^{n-k}}{\binom{n}{w}2^w} + 3^{n/2-k_2}\mathbb{P}(\mathcal{E}_{2,1})\right) \\
&= \frac{1}{3^{n-k}}\left(1 + \frac{3^{n-k}}{\binom{n}{w}2^w} + \sum_{t=0}^{n/2} \frac{3^{n/2-k_2}}{\binom{n/2}{t}2^t\binom{n}{w}^22^{2w}}\left(\sum_{\substack{i=\max\{0, 2t-w\} \\ w+i \equiv 0 \pmod{2}}}^t \binom{n/2}{i}\binom{n/2-i}{t-i}\binom{n/2-t}{(w+i)/2-t}2^{\frac{w+3i}{2}}\right)^2\right).
\end{aligned} \tag{19}$$

Thus, we have:

$$\mathbb{P}(\mathbf{H}_{pk}\mathbf{x}^\top = \mathbf{H}_{pk}\mathbf{y}^\top) = \frac{1}{3^{n-k}}(1 + \theta), \tag{20}$$

where

$$\theta \leq \frac{3^{n-k}}{\binom{n}{w}2^w} + \sum_{t=0}^{n/2} \frac{3^{n/2-k_2}}{\binom{n/2}{t}2^t\binom{n}{w}^22^{2w}}\left(\sum_{\substack{i=\max\{0, 2t-w\} \\ w+i \equiv 0 \pmod{2}}}^t \binom{n/2}{i}\binom{n/2-i}{t-i}\binom{n/2-t}{(w+i)/2-t}2^{\frac{w+3i}{2}}\right)^2.$$

Finally, by applying Lemma 1, we complete the proof of Theorem 4.  $\square$

### 3.2.4 EUF-CMA Security

With the previous preparations, we now prove that our proposed UVW signature scheme is EUF-CMA secure in the random oracle model.

**Theorem 5.** *Assume that the generalized  $(U + V, U + W)$  code distinguishability problem and the DOOM problem are  $(\tau_{GDP}, \epsilon_{GDP})$ - and  $(\tau_{DOOM}, \epsilon_{DOOM})$ -secure, respectively. Then, the UVW signature scheme is  $(\tau, \epsilon, Q_{\mathcal{H}}, Q_{\Sigma})$ -EUF-CMA secure in the random oracle model, where  $\tau$  and  $\epsilon$  are defined as follows:*

$$\epsilon \leq \frac{2^{\lambda_0}}{2^{\lambda_0} - Q_{\mathcal{H}}} \epsilon_{DOOM} + \epsilon_{GDP} + \frac{Q_{\Sigma}}{2} \sqrt{\theta}, \quad \tau \geq \tau_{DOOM} - (Q_{\mathcal{H}} + Q_{\Sigma})O(n(n-k)). \tag{21}$$

Here,  $O(n(n-k))$  is the time required to compute a syndrome for an  $[n, k]$  linear code.

**Proof.** Let  $\mathcal{A}$  be a  $(\tau, \epsilon, Q_{\mathcal{H}}, Q_{\Sigma})$ -adversary against the UVW signature scheme in the EUF-CMA model, and let  $(\mathbf{H}_0, \mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}})$  be a random instance of the DOOM problem with parameters  $n, k, Q_{\mathcal{H}}$ , and  $w$ . We emphasize that the syndromes  $\mathbf{s}_j$  are independent and uniformly distributed

random vectors in the vector space  $\mathbb{F}_3^{n-k}$ . Let **Game 0** (Figure 3) be the standard EUF-CMA game for the UVW scheme. We use  $\mathbb{P}(S_i)$  to denote the probability that the adversary  $\mathcal{A}$  wins **Game i**. Then we have:

$$\mathbb{P}(S_0) = \text{Succ}_S^{\text{EUF-CMA}}(\mathcal{A}). \quad (22)$$

**Input:** Adversary  $\mathcal{A}$

- 1**  $(\mathbf{H}_{\text{sk}}, \mathbf{H}_{\text{pk}}) \leftarrow \text{KeyGen}(1^\lambda)$
- 2** Setup random oracles  $\mathcal{H}$  and  $\Sigma$
- 3**  $(\mathbf{m}', \mathbf{e}', \mathbf{r}') \leftarrow \mathcal{A}^{\mathcal{H}, \Sigma}(\mathbf{H}_{\text{pk}})$
- 4** **If**  $\mathbf{e}' \mathbf{H}_{\text{pk}}^\top = \mathcal{H}(\mathbf{m}', \mathbf{r}')$ ,  $|\mathbf{e}'| = w$  and  $\Sigma$  has never output  $\mathbf{e}'$  **then**
- 5**  $\mathcal{A}$  wins the game
- 6** **else**
- 7**  $\mathcal{A}$  loses the game

Figure 3: **Game 0:** Standard EUF-CMA game for UVW

For the sake of the proof, we define three lists:  $\Lambda_{\mathcal{H}}$ ,  $\Lambda_{\Sigma}$ , and  $\Lambda$ . The oracles  $\mathcal{H}$  and  $\Sigma$  maintain the query lists  $\Lambda_{\mathcal{H}}$  and  $\Lambda_{\Sigma}$  with their respective output values. Since the oracles are controlled by the challenger, each simulated oracle can access these lists. List  $\Lambda_{\mathcal{H}}$  stores syndromes  $\mathbf{s}$  and their corresponding large-weight vectors  $\mathbf{e}$ ; that is, for any message  $\mathbf{m}$  and random vector  $\mathbf{r}$ ,  $\Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r}) = (\mathbf{s}, \mathbf{e})$ . List  $\Lambda_{\Sigma}$  stores the signature  $\sigma$  of message  $\mathbf{m}$ , i.e.,  $\Lambda_{\Sigma}(\mathbf{m}) = (\mathbf{e}, \mathbf{r})$ . List  $\Lambda$  stores the random vector  $\mathbf{r}$  corresponding to message  $\mathbf{m}$ , i.e.,  $\Lambda(\mathbf{m}) = \mathbf{r}$ . If there is no value associated with a query in the list, we use  $\perp$  to denote the output.

**Game 1.** In this game, the challenger replaces the hash oracle  $\mathcal{H}$  with a simulated oracle  $\mathcal{H}'$  (Figure 4). For any query  $(\mathbf{m}, \mathbf{r})$ , we observe two cases: either  $\mathbf{r} \neq \Lambda(\mathbf{m})$  or  $\mathbf{r} = \Lambda(\mathbf{m})$ . When  $\mathbf{r} \neq \Lambda(\mathbf{m})$ , the behavior of  $\mathcal{H}'$  is identical to that of  $\mathcal{H}$ . When  $\mathbf{r} = \Lambda(\mathbf{m})$ ,  $\mathcal{H}'$  chooses a vector  $\mathbf{e}$  uniformly at random from the set  $\mathcal{E}_{n,w}$ , computes  $\mathbf{s} = \mathbf{e} \mathbf{H}_{\text{pk}}^\top$  as the syndrome, and finally stores  $(\mathbf{s}, \mathbf{e})$  in the list  $\Lambda_{\mathcal{H}}$ . By the end of the simulation, the oracle  $\mathcal{H}'$  has generated a total of  $Q_{\mathcal{H}} + Q_{\Sigma}$  syndromes, some of which originate from the modified part of oracle  $\mathcal{H}$ . Let  $X$  be a random variable denoting the number of syndromes generated by the modified part of the oracle. Therefore, the probability that the adversary  $\mathcal{A}$  wins **Game 1** is as follows:

$$\begin{aligned} \mathbb{P}(S_1) &= \mathbb{P}[(S_1 \cap (X = 0)) \cup (S_1 \cap (X > 0))] \\ &= \mathbb{P}[S_1 \cap (X = 0)] + \mathbb{P}[S_1 \cap (X > 0)]. \end{aligned} \quad (23)$$

Obviously,  $\mathbb{P}[S_1 \cap (X = 0)]$  represents the probability that the adversary  $\mathcal{A}$  wins the game when all syndromes are generated by the random oracle  $\mathcal{H}$ , which is exactly the probability that  $\mathcal{A}$  wins **Game 0**. Thus, we have:

$$\mathbb{P}(S_0) \leq \mathbb{P}(S_1). \quad (24)$$

```

Input: A pair  $(\mathbf{m}, \mathbf{r})$ 
Output: Syndrome  $\mathbf{s}$ 
1 If  $\Lambda(\mathbf{m}) = \perp$  then
2    $\Lambda(\mathbf{m}) \xleftarrow{\$} \{0, 1\}^{\lambda_0}$ 
3  $(\mathbf{s}, \mathbf{e}) \leftarrow \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r})$ 
4 If  $\mathbf{r} \neq \Lambda(\mathbf{m})$  then
5   If  $\mathbf{s} = \perp$  then
6      $\mathbf{s} \xleftarrow{\$} \mathbb{F}_3^{n-k}, \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r}) \leftarrow (\mathbf{s}, \perp)$ 
7 else
8   If  $\mathbf{s} = \perp$  then
9      $\mathbf{e} \xleftarrow{\$} \mathcal{E}_{n,w}, \mathbf{s} \leftarrow \mathbf{eH}_{\text{pk}}^T, \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r}) \leftarrow (\mathbf{s}, \mathbf{e})$ 
10 Return  $\mathcal{H}'(\mathbf{m}, \mathbf{r}) = \mathbf{s}$ 

```

Figure 4:  $\mathcal{H}'$ : Simulation of  $\mathcal{H}$  (**Game 1**)

**Game 2.** In this game, the challenger replaces  $\mathcal{H}'$  with  $\mathcal{H}''$  (Figure 5). The only difference between  $\mathcal{H}''$  and  $\mathcal{H}'$  lies in the generation method of the syndrome in Line 6. The oracle  $\mathcal{H}''$  sequentially takes the syndromes from the DOOM problem instance as the hash outputs. As emphasized earlier, the  $\mathbf{s}_j$ 's are independent and uniformly distributed random vectors in the vector space  $\mathbb{F}_3^{n-k}$ , hence the simulation remains completely indistinguishable, yielding:

$$\mathbb{P}(S_2) = \mathbb{P}(S_1). \quad (25)$$

```

Input: A pair  $(\mathbf{m}, \mathbf{r})$ 
Output: Syndrome  $\mathbf{s}$ 
1 If  $\Lambda(\mathbf{m}) = \perp$  then
2    $\Lambda(\mathbf{m}) \xleftarrow{\$} \{0, 1\}^{\lambda_0}$ 
3  $(\mathbf{s}, \mathbf{e}) \leftarrow \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r})$ 
4 If  $\mathbf{r} \neq \Lambda(\mathbf{m})$  then
5   If  $\mathbf{s} = \perp$  then
6      $\mathbf{s} \leftarrow \text{Next}.[\mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}}], \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r}) \leftarrow (\mathbf{s}, \perp)$ 
7 else
8   If  $\mathbf{s} = \perp$  then
9      $\mathbf{e} \xleftarrow{\$} \mathcal{E}_{n,w}, \mathbf{s} \leftarrow \mathbf{eH}_{\text{pk}}^T, \Lambda_{\mathcal{H}}(\mathbf{m}, \mathbf{r}) \leftarrow (\mathbf{s}, \mathbf{e})$ 
10 Return  $\mathcal{H}''(\mathbf{m}, \mathbf{r}) = \mathbf{s}$ 

```

Figure 5:  $\mathcal{H}''$ : Simulation of  $\mathcal{H}'$  (**Game 2**)

**Game 3.** In this game, the challenger replaces the signature oracle  $\Sigma$  with a simulated oracle  $\Sigma'$  (Figure 6). Since  $\Sigma'$  queries  $\mathcal{H}''$  with  $(\mathbf{m}, \Lambda(\mathbf{m}))$  and stores the syndrome and its corresponding large-weight vector in the list  $\Lambda_{\mathcal{H}}$ , it does not require the private key to generate a signature. Line

5 in  $\Sigma'$  ensures that two distinct queries for the same message will generate different signatures. The only difference between **Game 3** and **Game 2** is the method of generating syndromes during signature queries. While **Game 2** selects them uniformly at random from the vector space  $\mathbb{F}_3^{n-k}$ , **Game 3** computes  $\mathbf{e}\mathbf{H}_{\text{pk}}^\top$  as the syndrome, where  $\mathbf{e} \xleftarrow{\$} \mathcal{E}_{n,w}$ . Utilizing Proposition 4 and Lemma 4 from reference [5], we have:

$$|\mathbb{P}(S_2) - \mathbb{P}(S_3)| \leq \frac{Q_\Sigma}{2} \sqrt{\theta}, \quad (26)$$

where  $\theta$  is given in Theorem 4.

**Input:** Message  $\mathbf{m}$   
**Output:** Signature  $(\mathbf{e}, \mathbf{r})$   
**1** if  $\Lambda(\mathbf{m}) = \perp$  then  
**2**     $\Lambda(\mathbf{m}) \xleftarrow{\$} \{0, 1\}^{\lambda_0}$   
**3**  $\mathcal{H}''(\mathbf{m}, \Lambda(\mathbf{m}))$   
**4**  $(\mathbf{s}, \mathbf{e}) \leftarrow \Lambda_{\mathcal{H}}(\mathbf{m}, \Lambda(\mathbf{m}))$   
**5**  $\Lambda(\mathbf{m}) \leftarrow \perp$   
**6** return  $\Lambda_\Sigma(\mathbf{m}) = (\mathbf{e}, \mathbf{r})$

Figure 6:  $\Sigma'$ : Simulation of  $\Sigma$  (**Game 3**)

**Game 4.** In this game, the public key matrix  $\mathbf{H}_{\text{pk}}$  is replaced by the matrix  $\mathbf{H}_0$  from the DOOM problem instance  $(\mathbf{H}_0, \mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}})$ . To establish a connection between **Game 4** and **Game 3**, we construct a distinguisher  $\mathcal{D}$ . If  $\mathbf{H}$  is generated by the key generation algorithm KeyGen, then  $\mathcal{D}$  is simulating **Game 3**; thus, we have:

$$\mathbb{P}[\mathbf{H} \xleftarrow{\$} \mathcal{H}_{\text{pk}}(n, k) : \mathcal{D}(\mathbf{H}) = 1] = \mathbb{P}(S_3).$$

If  $\mathbf{H}$  is the matrix  $\mathbf{H}_0$  from the instance  $(\mathbf{H}_0, \mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}})$ , then  $\mathcal{D}$  is simulating **Game 4**; thus, we have:

$$\mathbb{P}[\mathbf{H} \leftarrow \mathbf{H}_0 : \mathcal{D}(\mathbf{H}) = 1] = \mathbb{P}(S_4).$$

Therefore, the advantage of the distinguisher  $\mathcal{D}$  is as follows:

$$\text{Adv}^{GDP}(\mathcal{D}) = |\mathbb{P}(S_3) - \mathbb{P}(S_4)|.$$

Furthermore, since the generalized  $(U + V, U + W)$  code distinguishability problem is  $(\tau_{GDP}, \epsilon_{GDP})$ -secure, we obtain:

$$|\mathbb{P}(S_3) - \mathbb{P}(S_4)| \leq \epsilon_{GDP}. \quad (27)$$

**Input:** Parity-check matrix  $\mathbf{H}$   
**Output:** A bit  $b$

- 1 Setup oracles  $\mathcal{H}''$  and  $\Sigma'$
- 2  $(\mathbf{m}', \mathbf{e}', \mathbf{r}') \leftarrow \mathcal{A}^{\mathcal{H}'', \Sigma'}(\mathbf{H})$
- 3 if  $\mathbf{e}'\mathbf{H}^\top = \mathcal{H}''(\mathbf{m}', \mathbf{r}')$ ,  $|\mathbf{e}'| = w$  and  $\Sigma'$  has never output  $\mathbf{e}'$   
**then**
- 4     Output 1
- 5 **else**
- 6     Output 0

Figure 7:  $\mathcal{D}(\mathbf{H})$  (Game 4)

We assume that the adversary  $\mathcal{A}$  outputs a valid signature  $(\mathbf{e}', \mathbf{r}')$  for message  $\mathbf{m}'$ . Let  $p$  be the probability that  $\mathbf{e}'\mathbf{H}_0^\top \in \{\mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}}\}$ , then we have:

$$p \geq (1 - \frac{1}{2^{\lambda_0}})^{Q_{\mathcal{H}}} \geq 1 - \frac{Q_{\mathcal{H}}}{2^{\lambda_0}}, \quad (28)$$

That is,

$$\text{Succ}^{\text{DOOM}}(\mathcal{A}) = p \cdot \mathbb{P}(S_4) \geq (1 - \frac{Q_{\mathcal{H}}}{2^{\lambda_0}})\mathbb{P}(S_4). \quad (29)$$

Since the DOOM problem is  $(\tau_{\text{DOOM}}, \epsilon_{\text{DOOM}})$ -secure, we obtain:

$$\text{Succ}^{\text{DOOM}}(\mathcal{A}) \leq \epsilon_{\text{DOOM}}. \quad (30)$$

Based on Equations (22), (24), (25), (26), (27), (29), and (30), we deduce:

$$\text{Succ}_S^{\text{EUF-CMA}}(\mathcal{A}) \leq \frac{2^{\lambda_0}}{2^{\lambda_0} - Q_{\mathcal{H}}} \epsilon_{\text{DOOM}} + \epsilon_{\text{GDP}} + \frac{Q_{\Sigma}}{2} \sqrt{\theta}.$$

The running time for successfully solving the DOOM instance  $(\mathbf{H}_0, \mathbf{s}_1, \dots, \mathbf{s}_{Q_{\mathcal{H}}})$  is the sum of the adversary  $\mathcal{A}$ 's running time and the time to compute  $Q_{\mathcal{H}} + Q_{\Sigma}$  hash values. The time required to compute these hash values is at most the time to compute  $Q_{\mathcal{H}} + Q_{\Sigma}$  syndromes. This yields the formula for  $\tau$ .

□

### 3.3 Concrete Security

The security of the UVW signature scheme is based on two core hardness assumptions: 1) the DOOM problem (Problem 2); 2) distinguishing whether a given linear code is a generalized  $(U + V, U + W)$  code. Against these two assumptions, there are two main attack paths:

1. **Forgery Attack (Message Attack):** Aims to solve the large-weight syndrome decoding problem over the ternary field. Namely, given the public key  $\mathbf{H}$  and the syndrome  $\mathbf{s}$ , the goal is to find an error vector  $\mathbf{e}$  satisfying  $\mathbf{e}\mathbf{H}^\top = \mathbf{s}$  and  $wt(\mathbf{e}) = w$ .
2. **Key Recovery Attack (Structural Attack):** Aims to distinguish the generalized  $(U + V, U + W)$  code from a random code using the structural characteristics of the code, thereby recovering the private key structure.

### 3.3.1 Key Recovery Attack and Codeword Distribution Analysis

The key recovery attack exploits the structural weakness hidden in the generator matrix of the UVW code. When  $\mathbf{v} = \mathbf{w} = \mathbf{0}$ , the codeword takes the form  $(\mathbf{u}||\mathbf{u})\mathbf{D}$ . For a specific weight  $w$ , the statistical properties of such codewords may deviate significantly from those of random codes, thus serving as a basis for the distinguisher. Proposition 6 gives the codeword weight distribution of a generalized  $[n, k]$   $(U + V, U + W)$  code over the finite field  $\mathbb{F}_3$ .

**Proposition 6.** *Let  $\mathcal{C}$  be a generalized  $[n, k]$   $(U + V, U + W)$  code over the finite field  $\mathbb{F}_3$ , where  $U$  is a random  $[\frac{n}{2}, k_1]$  linear code, and both  $V$  and  $W$  are random  $[\frac{n}{2}, k_2]$  linear codes. Let  $N_{(\mathbf{u}, \mathbf{u})}$  and  $N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})}$  be the average number of codewords of weight  $w$  taking the forms  $(\mathbf{u}||\mathbf{u})\mathbf{D}$  and  $(\mathbf{u} + \mathbf{v}||\mathbf{u} + \mathbf{w})\mathbf{D}$  (where  $\mathbf{v} \neq \mathbf{0}, \mathbf{w} \neq \mathbf{0}$ ), respectively. When  $w$  is even, we have:*

$$N_{(\mathbf{u}, \mathbf{u})}(w) = \frac{\binom{n/2}{w/2} 2^{w/2}}{3^{n/2-k_1}}, \quad N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})}(w) = \frac{\binom{n}{w} 2^w}{3^{n-k}} \left(1 - \frac{1}{3^{k_2}}\right),$$

and when  $w$  is odd, we have:

$$N_{(\mathbf{u}, \mathbf{u})}(w) = 0, \quad N_{(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{w})}(w) = \frac{\binom{n}{w} 2^w}{3^{n-k}} \left(1 - \frac{1}{3^{k_2}}\right).$$

At the parameter design level, to achieve an optimal balance between signature size and public key size, and to adhere to the structural properties of the Wave signature construction, this scheme first fixes the code rate  $R = k/n = 0.5$  (i.e.,  $k = n/2$ ). Based on this, considering that the trapdoor decoding capacity is determined by the dimension  $k_1$ , while the message unforgeability depends on the signature weight  $w$ , there is a tight coupling between these two parameters. To maximize security, we perform a joint parameter optimization on  $(k_1, w)$ , setting the working point at  $w = 0.89n$  and  $k_1 \gtrsim 0.67k$ . This setting ensures the success rate of the signature algorithm while simultaneously maximizing the difficulty of message forgery (high  $w$ ) and minimizing the leakage of the public key structure (low  $k_1$ ) (detailed complexity analysis can be found in Section 3.3).

Under this setting, based on Proposition 6, we plot the relationship between the average number of these two types of codewords and the relative weight  $x = \frac{w}{n}$  (as shown in Figure 8). It can be observed that when the relative weight  $w/n$  is less than 0.25 or greater than 0.991, the number of codewords with the structure  $(\mathbf{u}||\mathbf{u})\mathbf{D}$  possesses an overwhelming advantage. This suggests that key recovery attacks are highly likely to use codewords at these distribution edges as entry points; therefore, the ISD algorithm must be used to evaluate their concrete complexity.

To evaluate the concrete security bits of this scheme in both classical and quantum computing environments, we adopt the latest analysis framework proposed by Loyer et al. [17]. Although the UVW scheme is constructed based on  $(U + V, U + W)$  codes, differing in details from the  $(U, U + V)$  structure of the Wave scheme, the two are essentially isomorphic in security: forgery attacks depend solely on the parameters  $(n, k, w)$  and are independent of the specific generator matrix; meanwhile, the key recovery attack in UVW exploits the  $(\mathbf{u}||\mathbf{u})\mathbf{D}$  subspace structure, which is identical in principle to exploiting Type-U codewords in Wave. Therefore, the relevant analytical models are fully applicable.

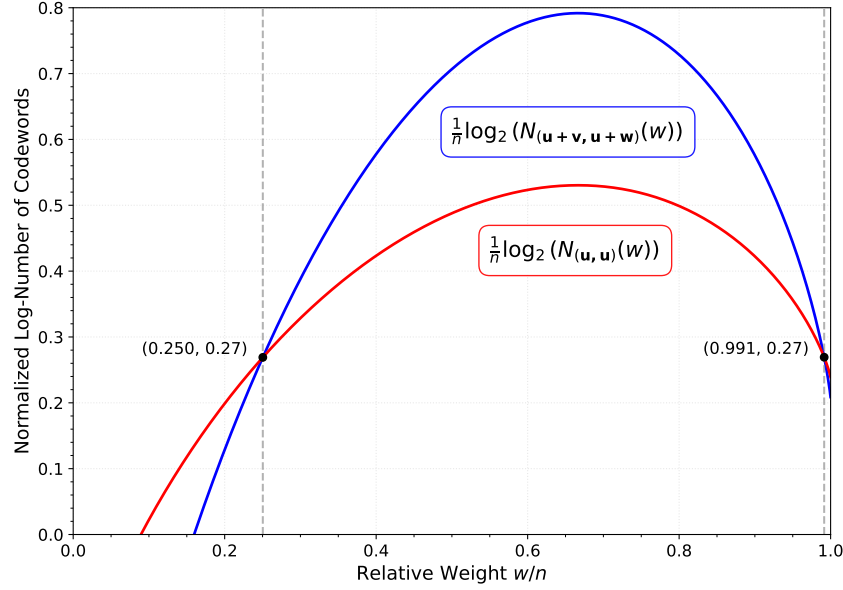


Figure 8: Average number of codewords of weight  $w$  taking the forms  $(\mathbf{u}||\mathbf{u})\mathbf{D}$  and  $(\mathbf{u} + \mathbf{v}||\mathbf{u} + \mathbf{w})\mathbf{D}$  (where  $\mathbf{v} \neq \mathbf{0}, \mathbf{w} \neq \mathbf{0}$ ) when  $k = 0.5n, k_1 \gtrsim 0.67k$

### 3.3.2 Attack Complexity Analysis

For forgery attacks, the attacker attempts to solve the DOOM problem. In the classical setting, the most efficient approach is to utilize the generalized birthday algorithm proposed by Wagner [18] and its smoothed variants tailored for the DOOM problem [19][20]. According to Theorem 7 by Loyer et al. [17], the complexity of the classical smoothed Wagner algorithm, denoted as  $T_{msg,class}$ , is:

$$T_{msg,class} = \max \left\{ \left( \frac{3^l}{2^{\frac{k+l}{2^{a-1}}}} \right)^{\frac{1}{a-2}}, \frac{3^{n-k-l}}{2^{w-p} \binom{n-k-l}{w-p}} \right\}$$

In the quantum setting, the attacker can combine the quantum amplitude amplification technique proposed by Brassard et al. [21] with quantum list merging to achieve acceleration. According to Theorem 8 by Loyer et al. [17], its quantum complexity  $T_{msg,quantum}$  is:

$$T_{msg,quantum} = \max \left\{ \left( \frac{3^l}{2^{\frac{k+l}{2^{a-1}}}} \right)^{\frac{1}{a-2}}, \sqrt{\frac{3^{n-k-l}}{2^{w-p} \binom{n-k-l}{w-p}}} \right\}$$

where  $a$  is the depth of the algorithm's recursion tree, and  $p, l$  are optimization parameters. Through comparison, it can be observed that the quantum algorithm primarily provides a square-root level speedup for the second term (the solution search part) in the formula. This complexity is insensitive to the code structure parameter  $k_1$  and is mainly determined by  $(n, k, w)$ .

For key recovery attacks, the attacker attempts to find a special codeword that falls into the region dominated by the red curve distribution in Figure 8. The most effective classical attack

employs the collision search algorithm proposed by Dumer [8] in 1989 and its derivative variants. Derived from Theorem 5 by Loyer et al. [17], the classical attack complexity  $T_{key,class}$  is:

$$T_{key,class} = \tilde{O} \left( \max \left\{ \sqrt{\binom{k+l}{p} 2^p}, \frac{3^{n/2-k_1} \sqrt{\binom{n}{t}}}{2^{(t-p)/2} \sqrt{\binom{k+l}{p} \binom{n-k-l}{t-p}}} \right\} \right)$$

In the quantum setting, ISD algorithms utilize the quantum walk model proposed by Kachigar and Tillich [22] to further accelerate the search process. According to Theorem 6 by Loyer et al. [17], the corresponding lower bound for the quantum complexity is:

$$T_{key,quantum} = \tilde{O} \left( \max \left\{ \left( \binom{k+l}{p} 2^p \right)^{1/3}, \frac{3^{n/4-k_1/2} \left( \binom{n}{t} \right)^{1/4}}{2^{t/4-p/2} \sqrt{\binom{n-k-l}{t-p}}} \right\} \right)$$

where  $t$  is the weight of the target  $(\mathbf{u}||\mathbf{u})\mathbf{D}$  codeword, and  $p, l$  are the optimization parameters of the ISD algorithm. Comparing the two equations reveals that the quantum algorithm provides polynomial-level speedup in both list searching and solution verification.

### 3.3.3 Parameter Optimization and Overall Security Evaluation

To determine the theoretical security trend of the parameter ratios, we conducted a quantitative analysis using the first set of parameters (Level 1:  $n = 9700, k = 4850, w = 8633$ ) as a baseline. Figure 9 illustrates the trend of quantum security varying with the  $k_1/k$  ratio under this specification.

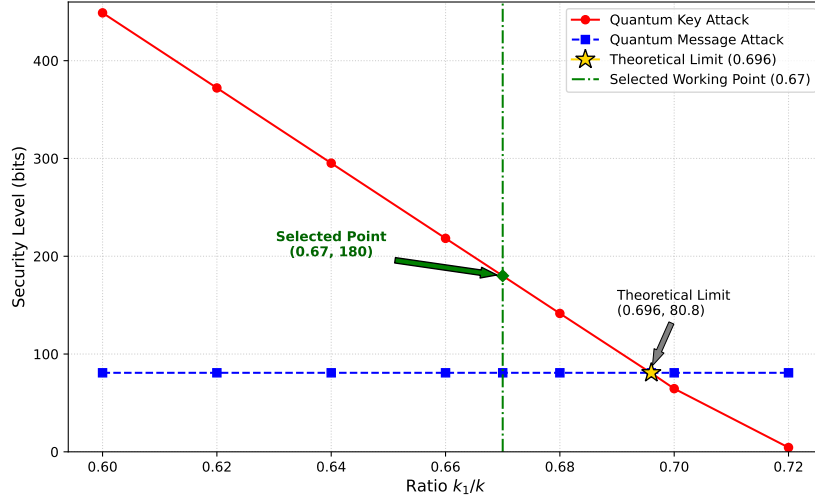


Figure 9: Trend of quantum security varying with the ratio  $k_1/k$  (based on  $n = 9700, k = 4850, w = 8633$ )

The analysis shows that the key recovery attack complexity (solid line) decreases as  $k_1/k$  increases, while the forgery attack complexity (dashed line) remains stable. The two lines intersect

at  $k_1/k \approx 0.696$ , which is precisely the “max-min security balance point” defined by Sendrier in the Wave scheme.

As mentioned in the 2 section, although  $k_1/k \approx 0.696$  is the theoretical balance point, this scheme ultimately selects  $k_1/k \approx 0.67$  as the actual working point out of considerations for distribution matching and resistance to side-channel attacks. As clearly seen in Figure 9, decreasing  $k_1$  significantly enhances key security, providing greater security redundancy for the system to defend against structural recovery attacks.

It should be pointed out that although Figure 9 is plotted based on the first set of parameters (Level 1), the conclusion regarding this optimal ratio is equally applicable to the other parameter sets of this scheme. This is because the attack complexity of this scheme primarily depends on relative metrics such as the code rate ( $R = k/n$ ) and the relative weight ( $w/n$ ), rather than the absolute code length. Across all security levels defined in this scheme, we maintained a uniform structural ratio (i.e., fixing  $k = 0.5n$  and  $w = 0.89n$ ). The numerical verification results also indicate that as  $n$  increases (Level 2 and Level 3), the  $k_1/k \approx 0.67$  ratio strategy remains valid. Accordingly, we calculated the concrete security bits for the three recommended parameter sets (Level 1, 2, and 3) under the current best-known cryptanalytic methods, as shown in Table 2.

Table 2: Concrete Security Estimates for the UVW Signature Scheme

| Scheme           | $(q, n, k, k_1, k_2, w)$              | Classical (Msg/Key) | Quantum (Msg/Key) |
|------------------|---------------------------------------|---------------------|-------------------|
| UVW128 (Level 1) | (3, 9700, 4850, 3250, 1600, 8633)     | 133.7 / 304.5       | 80.8 / 179.5      |
| UVW256 (Level 2) | (3, 19200, 9600, 6433, 3167, 17088)   | 257.7 / 603.1       | 156.6 / 352.7     |
| UVW512 (Level 3) | (3, 39000, 19500, 13066, 6434, 34710) | 517.4 / 1226.1      | 315.2 / 714.5     |

## 4 Performance Evaluation

We implemented the UVW signature scheme in C and provide two implementations: a reference implementation and an optimized implementation. The code for the reference implementation does not contain any platform-specific instruction set inlining (e.g., SSE/AVX intrinsics for x64) to ensure cross-platform portability. The optimized implementation targets mainstream 64-bit x64 processors and introduces SIMD (AVX2) optimizations for core operations, specifically including: SIMD implementations for vector addition, subtraction, and multiplication over  $\mathbb{F}_3$  (utilizing AVX2 256-bit vector instructions to process four 64-bit words simultaneously).

The experimental platform is as follows: CPU Intel(R) Core(TM) i9-14900K @ 3.20 GHz (24 CPU cores and 128 GB RAM), operating system Windows 11 (Version 23H2), and compiler GCC 15.2.0 (MSYS2 UCRT64). The reference implementation is compiled with the parameters `-DUSE_API_PKC -O2 -Wall -Wextra`; the optimized implementation is compiled with the parameters `-DUSE_API_PKC -O3 -march=native -mtune=native`. The comparisons in this paper are based on the results measured under the same testing platform and identical timing criteria.

Table 3 presents the actual storage sizes of the public key, private key, and signature for the UVW signature scheme under various parameter sets. Table 4 and Table 5 provide the CPU cycles and time required for key generation, signature generation, and verification of the UVW signature scheme using each parameter set, respectively. We utilize the `rdtsc` instruction and the `QueryPerformanceCounter` function to acquire the CPU cycles and time, running each algorithm for every parameter set 100 times to calculate the average elapsed values.

Table 3: Actual Storage Sizes of Keys and Signatures in the UVW Signature Scheme (Bytes)

| Parameter Set    | Private Key Size | Public Key Size | Signature Size |
|------------------|------------------|-----------------|----------------|
| UVW128 (Level 1) | 40               | 5897612         | 1244           |
| UVW256 (Level 2) | 40               | 23040012        | 2444           |
| UVW512 (Level 3) | 72               | 95160012        | 4956           |

Table 4: Required CPU Cycles for Key Generation, Signature, and Verification in the UVW Signature Scheme (in  $10^6$  cycles)

| Parameter Set    | Implementation | Key Generation | Signature | Verification |
|------------------|----------------|----------------|-----------|--------------|
| UVW128 (Level 1) | Reference      | 34041.21       | 11164.84  | 207.91       |
|                  | Optimized      | 21472.33       | 6693.16   | 193.67       |
| UVW256 (Level 2) | Reference      | 581621.32      | 132446.21 | 1361.49      |
|                  | Optimized      | 239723.46      | 74971.41  | 1331.37      |
| UVW512 (Level 3) | Reference      | 3412692.86     | 545418.51 | 6339.27      |
|                  | Optimized      | 2481804.09     | 384122.64 | 5638.58      |

Table 5: Required Time for Key Generation, Signature, and Verification in the UVW Signature Scheme (in ms)

| Parameter Set    | Implementation | Key Generation | Signature | Verification |
|------------------|----------------|----------------|-----------|--------------|
| UVW128 (Level 1) | Reference      | 10832.66       | 3503.02   | 62.85        |
|                  | Optimized      | 6989.59        | 2100.15   | 60.75        |
| UVW256 (Level 2) | Reference      | 182486.44      | 41555.77  | 427.17       |
|                  | Optimized      | 75241.36       | 23522.63  | 417.72       |
| UVW512 (Level 3) | Reference      | 1070748.38     | 171127.08 | 1989.20      |
|                  | Optimized      | 778677.67      | 120520.28 | 1769.13      |

## 5 Features of the Scheme

The UVW signature scheme possesses the following prominent features in its design and implementation:

### 1. Algorithmic Innovation

This scheme utilizes generalized  $(U+V, U+W)$  codes as the core trapdoor for digital signatures. As demonstrated in the in-depth security analysis, this construction successfully expands the upper bound of the effective decoding interval of random linear codes to  $\frac{q-1}{q}n + \frac{2k_1}{q}$ .

### 2. Algorithmic Simplicity

The scheme follows the classical “Hash-and-Sign” paradigm, featuring a clear and compact logic. The underlying core signature operations primarily rely on the efficient Prange algorithm and basic linear algebra operations over the finite field  $\mathbb{F}_3$ . It avoids complex polynomial ring multiplications or Gaussian sampling, making the algorithm easy to understand, analyze, and deploy.

### 3. Algorithmic Flexibility and Scalability

The scheme framework exhibits strong scalability. By adjusting core parameters such as  $(q, n, k, k_1, k_2, w)$ , it can be smoothly extended to different security levels. Currently, the scheme natively provides three sets of recommended parameters (Level 1, Level 2, and Level 3), which perfectly match the classical security strength requirements of 128, 256, and 512 bits, respectively, as well as the quantum security strength requirements of no less than 80, 128, and 256 bits.

### 4. Algorithmic Compatibility

Regarding the choice of the hash function, this scheme employs the SM3 cryptographic hash algorithm for instantiation. This makes the scheme highly compatible with China’s existing commercial cryptographic infrastructure and standard systems, facilitating its integration and promotion in practical engineering applications.

### 5. Performance Advantages in Implementation Across Different Platforms

Extremely short signature size: Among code-based cryptographic schemes, this scheme achieves a highly competitive signature size. For instance, at the Level 1 security level, the signature size is only 1244 bytes.

Native resistance to side-channel attacks: Benefiting from the distribution-matching design of  $k_1/k \approx 0.67$ , the weight of the random part of the candidate signature is highly aligned with the target weight, significantly improving the single-sampling success rate. Combined with a hard-truncation mechanism tailored to the security level, the failure probability due to rejection sampling can be strictly controlled below the corresponding exponential lower bound (e.g., strictly less than  $2^{-128}$  for Level 1). This ensures that the time consumed for signature

generation is completely statistically decoupled from the private key information, naturally resisting timing side-channel attacks from the root of the algorithm design.

Advantages of platform instruction set acceleration: The underlying  $\mathbb{F}_3$  vector operations (addition, subtraction, multiplication) are highly friendly to SIMD architectures. On mainstream 64-bit processors, by introducing AVX2 (256-bit vector instructions) optimizations, four 64-bit words can be processed in parallel at a time, greatly increasing the computational throughput of signature generation and verification.

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